

An Arithmetic Family of Aperiodic Tilings in Odd Prime Dimensions

The CLARK Monotile and a Formal Proof in Five Dimensions

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Abstract

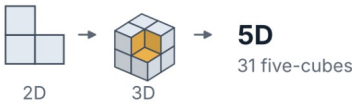
We give a uniform arithmetic construction of aperiodic chair matching rules in every odd prime dimension, yielding an explicit infinite family. In the general family, local matching rules for fully framed chairs on a common lattice force a hierarchy at arbitrarily large scales. We also construct an explicit five-dimensional aperiodic monotile, CLARK (Chair with Local Asymmetric Registration Keys): one shape that tiles space but permits no periodically repeating tiling. Its boundary alone enforces nonperiodicity, even when arbitrary translations, rotations and reflections are allowed.

A d -dimensional chair is a side-two cube with one corner unit cube omitted. The registered construction uses affine maps over $\mathbb{F}_p$; its tilings exist, have unique hierarchical decompositions and have finite Euclidean symmetry groups. A separate dense rational decoration gives physical monotiles with finite full tiling symmetry groups for primes $p \equiv 3 \pmod{4}$. CLARK has 31 unit five-dimensional cubes and 256 pyramidal keys. Its compactness, tiling existence and absence of every nonzero translation period are proved in Lean and have passed independent verification. Every tiling by CLARK has at most 3,840 full Euclidean symmetries. An exact seven-dimensional body with 127 cubes and 1,024 keys has a computer-assisted proof of tiling existence and absence of nonzero translation periods; its complete Lean verification remains open. The uniform results remain ordinary mathematical proofs.

CLARK: A Strongly Aperiodic Tile in Five Dimensions

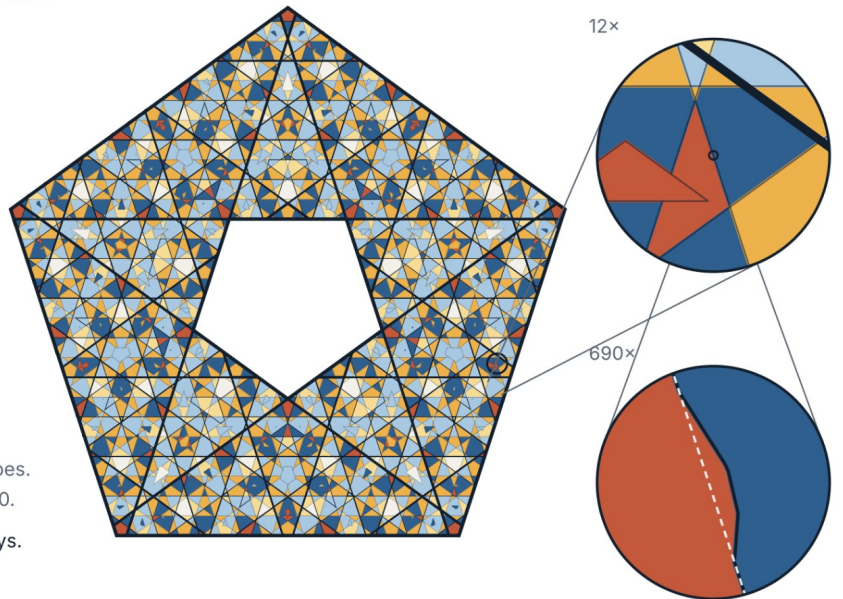
Copies of CLARK fill five-dimensional space, but no tiling repeats under a nonzero translation. Small bumps and sockets force the tiles into larger copies of the same chair, at every scale. Every tiling has a finite symmetry group.

The shape



Start with a five-dimensional cube of side two and remove one corner unit cube, leaving 31 unit five-cubes. Add 128 bumps and 128 sockets, each of height $1/240$.

CLARK: Chair with Local Asymmetric Registration Keys.



Aperiodicity independently verified. Finite-symmetry bound proved in Lean.

Exact planar slice of a level-5 CLARK supertile, keys included
0 1 2 3 4 5 reflected axes per tile; heavier lines, larger supertiles

One bump seated in its socket.
Dashed: the flat facet.

A section of a level-five CLARK carrier patch. The insets resolve matching boundary keys in the same plane, at approximately $12\times$ and $690\times$ the main scale. The plane and display conventions are specified in Appendix A.1.

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1 Results and scope

A nonempty proper subset of $\mathbb{F}_p$ has an invertible cardinality, so its mean can determine a coordinate permutation. The quadratic character of the permutation's slope cancels the parity of the reflected coordinates. These two facts produce proper child frames in every odd-prime dimension. Primality is sufficient for this construction; no necessity statement for aperiodic tilings is intended.

A registered matching rule specifies a common lattice, permissible coordinate frames and allowed contacts. A physical tile must enforce these constraints through its boundary. The following theorems distinguish the registered rule from the shapes that realize it.

Theorem 1.1 (Registered arithmetic hierarchy). *Let p be an odd prime, let $\mu \in \mathbb{F}_p^*$ be a square, and let $g \in \mathbb{F}_p^*$ be a nonsquare. Use the substitution of Section 2, including its stipulated empty role, and its least full-pose contact language E . Every full lattice-registered tiling by properly framed chairs obeying E has a unique nested partition into substituted parents at every scale. Such tilings exist. Every such framed tiling has no nonzero translation period and has a finite Euclidean symmetry group.*

This is an ordinary mathematical theorem for the specified full-frame model. It is not a claim that the undecorated chair, or a uniform family of sparse shapes in all odd-prime dimensions, enforces E .

Theorem 1.2 (Dense geometric family). *For each prime $p \equiv 3 \pmod{4}$, take $\mu = 1$, $g = -1$, and the full free-orbit rational facet construction T_p^{dense} of Section 6. Its registered full-profile mating relation is $\text{CL}(E)$, of cardinality*

$$2(2^p - 1) + 3p^2 + p.$$

The body is a rational polyhedral p -ball. It admits tilings by congruent copies, and every such tiling has a finite full Euclidean symmetry group. All Euclidean isometries are allowed in the definition of a tiling.

The proof combines registration in arbitrary dimension with the facet-closure argument for this prime family. It has not been fully formalized in Lean. The dense body differs from the following sparse bodies, including in dimension seven.

Theorem 1.3 (CLARK and HENRY). *Let the CLARK tile T_5 and the HENRY tile T_7 be the exact closed rational polyhedral bodies defined in Section 7 and by the coordinate files identified by SHA-256 hashes in Appendix A. Their respective data are*

	carrier cubes	exposed unit facets	keys	key height
CLARK (T_5)	31	160	256	1/240
T_7	127	896	1024	1/336.

Each body is compact, admits a tiling of its Euclidean space by arbitrary isometric copies, and every such tiling has no nonzero translation period of its physical tile collection.

The public exact coordinate specification, independent body and claim definition, and proof entry point expose the geometry and proposition being checked. The challenge file states the target; the solution supplies its proof. CLARK is also strongly aperiodic:

Theorem 1.4 (Full symmetry bound for CLARK). *Every tiling by arbitrary isometric copies of T_5 has at most $2^5 5! = 3,840$ Euclidean symmetries preserving its physical tile collection. The bound counts both orientation-preserving and orientation-reversing symmetries.*

The proof is given in Section 8.7. I use *strongly aperiodic* to mean that every admitted tiling has a finite full Euclidean symmetry group; the group need not be trivial.

Context. The planar hat and spectre constructions use hierarchical substitutions to prove aperiodicity [1, 2]. Tsiokos’s three-dimensional decorated chair supplies an antecedent for the passage from matching boundary features to lattice registration [4]. Kamenetsky develops a higher-dimensional full-frame contact formalism and a conditional certificate based on coarsening and local enclosure [3].

Goodman-Strauss proves that substitution hierarchies in dimension $d > 1$ can be enforced by finite matching rules under his hereditary-edge and sibling-edge-to-edge hypotheses [7]. Higher-dimensional chair substitutions also occur in the model-set literature: Lee and Moody treat the n -dimensional chair and its pure-point diffraction [10], while Baake and Grimm calculate diffraction for a weighted planar chair point set [11]. These results concern substitution systems and their structure; they do not identify the two sparse bodies considered here.

Socular and Taylor give a connected three-dimensional geometric realization of their matching rules, with tilings formed by slabs that stack periodically in the transverse direction [9, Section 7]. Greenfeld and Tao construct a single translational tile in sufficiently high dimension that admits tilings but no tiling periodic under a full-rank lattice [8, Remark 1.1 and Corollaries 1.6–1.7]. The absence of a full-rank period lattice does not by itself exclude a single nonzero period. The physical

statements here allow arbitrary isometries and exclude every nonzero translation period; the dense family and CLARK also have finite full symmetry groups.

Scherlis presents a September 2026 research note giving a construction and written proof of strongly aperiodic monotiles in every dimension $d \geq 2$ through keyed products of the Spectre and Chair44 [12]. He also reports exact and sampled searches for higher-dimensional face-decorated chairs. The exact search in dimensions four and five imposes an additional condition: the relative frame of the hypothetical siblings predicted by two adjacent tiles must depend only on their common notch axis (his virtual-pointer consistency). CLARK uses the standard chair dissection and identity frames for the central and empty children, but does not satisfy that full-frame condition. Its relevant relative frames are identified only after accounting for the body's symmetry R . The restricted search therefore does not exclude the construction given here.

The present work gives explicit arithmetic child frames and exact contact classifications for the registered prime-dimensional rule, separate geometric enforcement proofs for the dense and sparse constructions, and formal proofs for the exact five-dimensional tile, including its finite full-symmetry bound. Primality supplies the subset means and parity calculation used in this construction; it is not claimed to be necessary for aperiodicity.

Structure of the argument. The arithmetic sections prove the registered hierarchy and the separate dense realization. For CLARK, the geometric lemmas recover a lattice and exact contact relation from arbitrary isometric placements; the subsequent five-dimensional proof recognizes parents, constructs an infinite tiling and bounds its full symmetry group. Results obtained by exhaustive rational enumeration are labeled finite computations. Their complete tests and the formal verification records are given in the appendices.

2 The arithmetic substitution

2.1 Chairs and signed coordinate frames

Write $I = \mathbb{F}_p$, and identify subsets of I with their indicator vectors when they occur in vector formulas. The closed chair is

$$P_p = \bigcup_{A \subseteq I} (\mathbf{1}_A + [0, 1]^p). \quad (1)$$

The chair P_p is the *carrier* on which the later boundary features are placed. The union in (1) retains the exposed boundary of the omitted corner. Figure 1 shows four exact three-dimensional sections of CLARK. These sections avoid every key and therefore also show the corresponding carrier sections.

For a signed permutation $h = (f, s)$ I use the row action

$$(hx)_i = s_i x_{f(i)}, \quad D_h = \{i : s_i = -1\}.$$

Thus $f_{hk} = f_k \circ f_h$ and $D_{hk} = D_h \triangle f_h^{-1}(D_k)$. The frame is proper when $\det h = 1$. A registered pose (h, t) has $t \in \mathbb{Z}^p$. Its occupied cell lower corners, bounding-box lower corner, and omitted cell lower corner are respectively

$$t + hb - \mathbf{1}_{D_h}, \quad t - 2\mathbf{1}_{D_h}, \quad t + \mathbf{1} - 3\mathbf{1}_{D_h}, \quad (2)$$

where $b \in \{0, 1\}^p \setminus \{\mathbf{1}\}$. The correction $-\mathbf{1}_{D_h}$ is necessary because a signed linear map acts on a whole cube, not just on its lower corner.

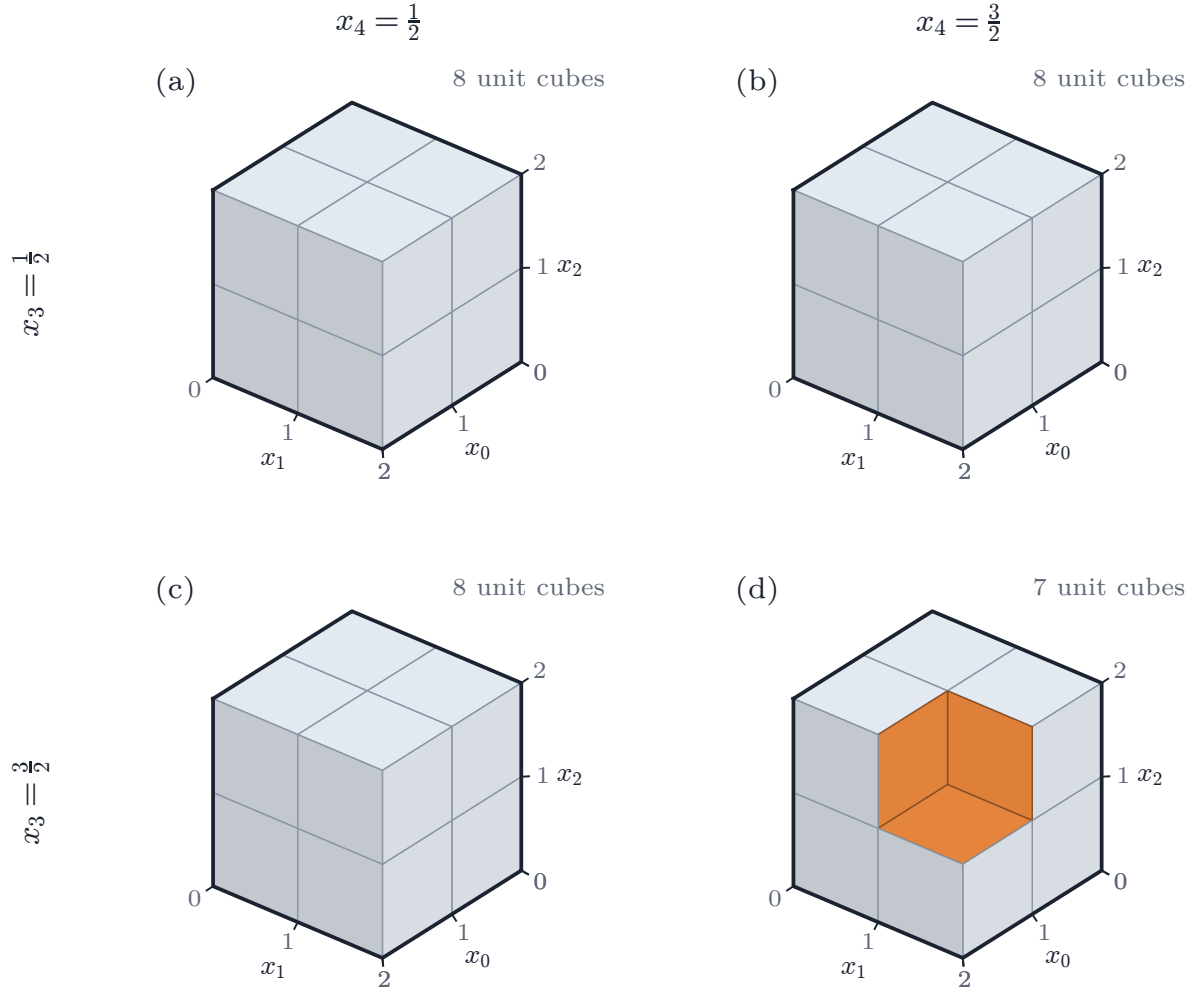


Figure 1: Exact sections of CLARK with (x_0, x_1, x_2) free and (x_3, x_4) fixed at the values shown. Panels (a)–(c) contain eight closed unit cubes; panel (d) contains seven. The three highlighted squares in (d) are the inward-facing boundary of the missing corner. None of the 256 key solids meets these section planes, so these sections of T_5 equal the carrier sections. The views are orthographic with equal coordinate scales.

For two poses, the normalized relative pose is

$$\text{rel}((H, t), (K, u)) = (H^{-1}K, H^{-1}(u - t)).$$

A face contact means an intersection of boundaries with positive $(p - 1)$ -dimensional measure. Contacts of lower dimension impose no rule here. A full framed tiling covers $\mathbb{R}^p$, has pairwise disjoint tile interiors, and retains each tile's complete pose as its marking.

2.2 The child poses

For $\emptyset \neq A \subsetneq I$, put

$$k = |A|, \quad z_A = k^{-1} \sum_{i \in A} i \in \mathbb{F}_p.$$

Choose a square $\mu \neq 0$ and a nonsquare g . The outer child for role A has pose (h_A, u_A) , where

$$f_A(i) = \mu g^{|A|}(i - z_A), \quad (h_A x)_i = (-1)^{\mathbf{1}_A(i)} x_{f_A(i)}, \quad u_A = 4\mathbf{1}_A. \quad (3)$$

For the empty outer role I stipulate $z_\emptyset = 0$ and $f_\emptyset(i) = \mu i$. The empty role is defined separately because its mean is undefined. There is no full outer role $A = I$. The remaining central child c has positive frame M_μ with index map $i \mapsto \mu i$ and translation $u_c = \mathbf{1}$.

Lemma 2.1 (Proper frames and exact dissection). *All these frames are proper, and the 2^p children partition $2P_p$ up to their boundaries. Refinement of a posed parent (H, t) gives the children*

$$(Hh_r, 2t + Hu_r).$$

Proof. The sign of the permutation $i \mapsto ai + b$ of $\mathbb{F}_p$ is the quadratic character $\chi(a)$, by Zolotarev's lemma for multiplication and the parity of translations [5, Corollary 11]. One direct proof applies the Vandermonde product to the permuted field elements: the sign is $a^{p(p-1)/2} = a^{(p-1)/2}$ in $\mathbb{F}_p$, hence $\chi(a)$. Translation is a p -cycle or the identity and is even. Therefore $\text{sgn}(f_A) = \chi(\mu g^{|A|}) = (-1)^{|A|}$, which cancels the product of the coordinate signs. The centre and empty child are proper because μ is a square.

Every coordinate permutation preserves the underlying chair. The outer child's occupied cells lie in the box $2A + \{0, 1\}^p$, with exactly the cell $\mathbf{1} + A$ missing. The outer boxes are disjoint and cover the doubled chair's macrocell boxes. The central child occupies exactly the cells $\mathbf{1} + A$ for proper A , filling precisely these holes. This proves the dissection. $\square$

2.3 The least contact language

Define E to be all normalized face contacts that occur inside some finite substituted supertile. Equivalently it is the least inverse-closed language containing all sibling contacts and closed under taking cross-child contacts. The symbolic catalog below proves that E is finite without enumerating the full signed-permutation group.

Let $\ell = \text{ord}(g)$ and write $g\langle g^2 \rangle$ for the odd powers of g . All contact permutations have affine slope in $\langle g \rangle$. There are at most $p\ell 2^{p-1}$ proper signed-affine relative frames. Individual child slopes involve μ , but it cancels from relative contacts; this bound is valid for every permitted μ .

3 The contact catalog

3.1 Contacts across an omitted cell

In a full registered carrier tiling the omitted unit cell of a tile T is occupied by exactly one other tile, denoted $O(T)$. It meets T across a unit facet. Let H be the following outgoing hole-owner catalog:

$$H_0 = (\text{id}, \mathbf{1}), \quad (4)$$

$$H(D, b) : \quad f(i) = g^{-|D|}i + b, \quad D_h = D, \quad t = \mathbf{1} + 2\mathbf{1}_D, \quad (5)$$

where $\emptyset \neq D \subsetneq I$, $\sum D = 0$, and $b \in I$.

Lemma 3.1. *The outgoing hole-owner contacts in E are exactly H , and $|H| = 2^p - 1$.*

Proof. Normalize an outer child A against the centre. Its sign set is $D = f_A(A)$, which is balanced, and its relative pose is (5) with $b = \mu z_A$. Conversely, given D, b , recover

$$A = \mu^{-1}(b + g^{-|D|}D).$$

This has mean $\mu^{-1}b$ and reconstructs the required sibling. The empty sibling gives H_0 .

There are no further hole types under refinement. An outer child's hole is filled by its centre sibling. For a central child whose hole is filled across a parent boundary, write the parent owner as (F, t) with $f(i) = ai + b$, sign set D , and $t = \mathbf{1} + 2\mathbf{1}_D$. Its outer role $A = f(D)$ supplies the relevant fine cell. After normalization at the root centre, the fine translation is $\mathbf{1}$ and the sign set is empty. For nonempty D , its index map is

$$i \mapsto ag^{|D|}i - \mu ag^{|D|}z_D.$$

For (5) this is the identity. For $D = \emptyset$, the separately defined empty role gives $i \mapsto ai + \mu b$, again the identity for H_0 . Thus each refinement of a central child's hole contact produces H_0 .

Translation acts freely on the k -element subsets of I for $1 \leq k < p$. Their means run once through I in each orbit, giving $\binom{p}{k}/p$ balanced subsets. Consequently

$$|H| = 1 + p \sum_{k=1}^{p-1} \frac{\binom{p}{k}}{p} = 2^p - 1.$$

□

3.2 Walls and exact generation

Every contact translation has constant coordinate parity. This holds for siblings and is preserved by the cross-child formula

$$w' = h_r^{-1}(2w + Ku_s - u_r),$$

since outer translations are even and the central translation is odd in every coordinate. A chair box with odd lower-corner parity can meet the root only with offsets ± 1 in every coordinate; its intersection box is one cell. A disjoint face contact is therefore a hole contact or its inverse. With even parity, the boxes must be adjacent on one axis, with lower corner $2\epsilon e_j$. These are *wall contacts*.

In addition to H and H^{-1} , use the following wall poses:

$$\begin{aligned}
P_+(j, b) : \quad & f(i) = gi + b, \quad D = \{j\}, \quad t = 4e_j, \\
& (j, b) = (0, 0) \text{ or } j \neq 0, \quad b \neq 0; \\
P_-(j, b) : \quad & f(i) = g^{-1}i + b, \quad D = \{j\}, \quad t = 4e_j, \\
& (j, b) = (0, 0) \text{ or } b \neq 0, \quad j \neq -gb; \\
M(a, j) : \quad & f(i) = ai, \quad D = \{j\}, \quad t = 0, \quad a \in g\langle g^2 \rangle; \\
B(b) : \quad & f(i) = i + b, \quad D = I \setminus \{0\}, \quad t = \mathbf{21}; \\
B(b)^{-1} : \quad & f(i) = i - b, \quad D = I \setminus \{b\}, \quad t = \mathbf{21} - 4e_b.
\end{aligned}$$

Proposition 3.2 (Exact contact language). *The union of these families with H and H^{-1} is exactly E . Every listed contact occurs by substitution level four. The set is independent of μ , and*

$$|E| = \begin{cases} 2^{p+1} + (p+1)^2 - 2, & \ell = 2, \\ 2^{p+1} + 2p^2 - 2p + 2 + p\ell/2, & \ell > 2. \end{cases} \quad (6)$$

Proof. Two outer children contact exactly when their roles differ in one coordinate. Addition to a nonempty target role of size k gives slope g and intercept $-gj/k$. At $j = 0$ this is zero; for $j \neq 0$, varying k realizes each nonzero intercept. For target size $k = 1$, use the empty source. For $2 \leq k \leq p-1$, translate the added point to zero and choose the source role of size $k-1$ in $I \setminus \{0\}$. Since $1 \leq k-1 \leq p-2$, exchanging two distinct elements shows that these source roles cannot all have sum zero. A source with nonzero sum, followed by scaling, gives the required nonzero axis. Inverting these addition contacts sends

$$(j, b) \mapsto (gj + b, -g^{-1}b),$$

which gives exactly the displayed P_- restrictions. The other siblings are the hole contacts already classified.

For closure, consider outer roles A and B across a parent wall with frame $f(i) = ai + b$, negative set D , and wall axis j . Put

$$C = A \triangle D \triangle \{j\}, \quad B = f(C).$$

The child wall has zero translation and a singleton negative set. In the row convention its index map is $f_B \circ f \circ f_A^{-1}$. With $z_\emptyset = 0$, direct composition gives

$$y \mapsto ag^{|C|-|A|}y + \mu g^{|C|}(az_A + b - z_{f(C)}). \quad (7)$$

For nonempty C , its affine image satisfies $z_{f(C)} = az_C + b$. For $C = \emptyset$, instead $z_{f(C)} = 0$ by the empty-role definition. Thus the intercept is $-\mu ag^{|C|}z_C$ when $A = \emptyset$ and $C \neq \emptyset$, and is $\mu(az_A + b)$ when $C = \emptyset$. In particular it is μb when both roles are empty. For nonempty A, C , the affine parameters reduce to

$$a' = ag^{|C|-|A|}, \quad b' = \mu ag^{|C|}(z_A - z_C). \quad (8)$$

For the P and M families, $C = A$, so the means cancel and the odd-power slope is retained. In the possible empty-empty case, the empty-role definition gives $b' = \mu b = 0$. For B and B^{-1} , $C = I \setminus A$ with both sets nonempty proper. Their means agree because $\sum I = 0$; the new slope is $g^{p-2|A|}$, an odd power.

For a parent hole contact the only new outer-outer wall uses $A = I \setminus \{j\}$ on the root side. Direct substitution gives slope one, intercept μj , negative set $I \setminus \{0\}$, and translation $\mathbf{21}$, hence

B ; reversing gives B^{-1} . Mixed central/outer contacts are hole contacts. Central/central children cannot create wall contacts. This proves closure.

The same calculation produces every B at level two. Refining a B with $|A| = k$ produces slopes g^{p-2k} , which exhaust the odd powers as $1 \leq k < p$. To obtain every axis for a fixed $M(a, j)$, refine using a singleton $A = \{x\}$ avoiding j , producing every nonzero axis, and a role $\{j+x, j-x\}$ with $x \neq 0$, producing axis zero. These roles are proper for $p \geq 3$. Thus the catalog is generated by level four.

For $\ell > 2$ the two P families have distinct slopes, each with $(p-1)^2 + 1$ elements; M has $p\ell/2$ elements and the two B families together have $2p$. When $\ell = 2$, $g = -1$ and the P union instead has $p^2 - p + 1$ elements. Together with the two directed hole families this gives (6). $\square$

4 Recognition of complete parents

A *complete face corona* of a tile consists of that tile and all its face neighbors in a full tiling; it covers every exposed unit facet with the actual neighbor on its other side. The local arguments below use simultaneous nonoverlap of the entire patch, rather than independent edge tests. A tile C uniquely determines a candidate star $S(C)$ with C central. Reversing the hole catalog shows that its possible incoming predecessors are exactly the $2^p - 1$ canonical outer children. A candidate complete parent containing T is therefore $S(T)$ or $S(O(T))$: if T is outer, that parent's centre must be the actual owner of its hole.

Normalize T to $(\text{id}, 0)$. Let I_A denote its canonical predecessor with omitted cell A , whose box has lower corner $2A - \mathbf{1}$. Let W_j^ε denote a possible tile in the adjacent same-parity box with lower corner $2\varepsilon e_j$. All boxes in a face-connected legal tiling occupy the two parity classes 0 and 1 modulo two. At most one tile occupies any one side-two box, because two chairs in that box overlap in at least $2^p - 2$ cells.

Lemma 4.1 (Completion from one nonempty predecessor). *If a complete corona of T contains one nonempty incoming predecessor I_A , it contains the whole star $S(T)$.*

Proof. For each occupied role A and axis j , the exterior cell

$$x(A, j) = A + (2A_j - 1)e_j$$

has exactly two possible owner boxes: I_A and $W_j^{2A_j-1}$. This follows coordinate by coordinate from the two box-parity classes. A tile in the first box must omit A , which is already occupied by T , so it is the catalog predecessor. Hence absence of I_A forces the corresponding wall to own $x(A, j)$.

The omitted cells of positive singleton, negative singleton, positive complement and negative complement walls are respectively

$$\mathbf{1} + e_j, \quad \mathbf{1} - 3e_j, \quad 3e_j, \quad -e_j.$$

For nonempty proper A , none equals $x(A, j)$. Thus if I_A is present, every wall $W_j^{2A_j-1}$ is absent, or two tiles would own this exterior cell. Any proper B sharing one coordinate bit with A now has I_B present. Only $B = I \setminus A$ shares none. Choose $j \in A$ and $k \notin A$. The proper nonempty role $\{j, k\}$ shares a bit with A and its complement and propagates presence to the last role. This uses $p \geq 3$. $\square$

Proposition 4.2 (Unique parent from two complete coronas). *If complete coronas of T and $O(T)$ are present, exactly one of $S(T)$ and $S(O(T))$ is complete.*

Proof. Suppose $S(T)$ is incomplete. Lemma 4.1 shows that its only possible incoming predecessor is $I_\emptyset$. All $2p$ same-grid walls are present: use a nonempty role on each side of each axis in the exterior-cell argument. If $I_\emptyset$ is present, the negative walls must omit $-e_j$, which is already occupied by that predecessor; hence they are negative complement walls.

Set $C = O(T)$. If T has a nonempty role in $S(C)$, Lemma 4.1 at C completes that star. Otherwise the contact is H_0 : C has frame identity and anchor $\mathbf{1}$, and T is its empty predecessor. If $S(C)$ were also incomplete, all its negative walls would be complement walls. In the coordinates of T , the j -th such wall has omitted cell $\mathbf{1} - e_j$, which T occupies. It is consequently a nonempty predecessor of T . Lemma 4.1 would complete $S(T)$, a contradiction.

To exclude both complete stars, normalize $S(O(T))$ to the standard doubled chair and let T have outer role A . The box of $S(T)$ has cell-coordinate intervals $[-1, 2]$ off A and $[1, 4]$ on A , with omitted block $\{1, 2\}^p$. Choose $j \notin A$ and $k \neq j$. Set $x_j = 0$, and set $x_k = 2$ if $k \notin A$, or $x_k = 1$ if $k \in A$; choose all other coordinates inside T 's box. This cell lies in both parents and outside both T and $O(T)$. The two subdivisions' noncentral children have opposite origin parities, so they cannot assign it to the same tile. Completeness would violate disjoint interiors. $\square$

Thus complete stars partition any full E -tiling: every tile lies in one, and two stars sharing a tile would be its two forbidden complete candidate parents. This proves first-level recognition. Iteration also needs a coarse legality theorem.

5 Coarse legality and the infinite hierarchy

Define $\text{CL}(E)$ by testing all disjoint touching doubled parents whose cross-child face contacts are in E , then dividing their relative translation by two when alignment is proved. Alignment is a conclusion, not an input to the test.

Lemma 5.1 (Exhaustive witnesses and alignment). *For every child-legal touching parent pair, a legal child contact (K, w) and roles r, s reconstruct its unscaled relative pose by*

$$F = h_r K h_s^{-1}, \quad v = u_r + h_r w - F u_s. \quad (9)$$

Every such v is even in all coordinates.

Proof. A positive-measure parent interface contains a unit panel away from lower-dimensional child boundaries. The children incident to that panel supply a contact witness, and solving the relative-pose equation gives (9). The formula and the parity invariant make v either all even or all odd. In the odd case, every constituent side-two cube of one parent has lower-corner parity opposite every constituent cube of the other. In each coordinate their intervals either overlap with positive length or are separated by a positive gap; they cannot merely meet at an endpoint. Any intersection of constituent cubes would therefore give interior overlap. Disjoint parents could not touch at all. Thus the odd case is impossible. $\square$

Proposition 5.2 (Exact coarse catalog). *The aligned relation $\text{CL}(E)$ consists of H , H^{-1} , and the following walls:*

- (i) $D = \{j\}$, $t = 4e_j$, $f(i) = ai + b$, with $a \in g\langle g^2 \rangle$ and any b ;
- (ii) $D = \{j\}$, $t = 0$, $f(i) = ai$, with $a \in g\langle g^2 \rangle$;
- (iii) $D = I \setminus \{j\}$, $t = \mathbf{21}$ or $\mathbf{21} - 4e_j$, $f(i) = ai + b$, with $a \in \langle g^2 \rangle$ and any b .

In particular,

$$|\text{CL}(E)| = 2(2^p - 1) + \frac{p\ell(3p + 1)}{2}. \quad (10)$$

Proof. First restrict the geometry. If parent boxes meet across one axis but have a nonzero transverse displacement, the incident outer-child wall has its contact axis and extra transverse axes negative. Every E wall has either just its contact axis or its complement negative; the former has no extra axis, and the latter omits the contact axis. Both possibilities are excluded. Thus disjoint boxes give canonical walls. If parent boxes overlap in 2^q coarse cells, their two holes can remove at most two, so $q \leq 1$. One overlap cell gives a hole contact. In the two-cell case the chairs own each other's holes; the incident refined outer children have all p axes negative, again impossible for an E wall. These fine interfaces exist because removing two hole panels leaves at least $2^{p-1} - 2 > 0$ panels.

For a coarse hole with sign set D , the central/outer fine contact has empty sign set and translation **1**. Its permutation is the carry map in Lemma 3.1, so it must be H_0 . For nonempty D , this forces slope $g^{-|D|}$ and $z_D = 0$; for empty D , the empty-role case forces the identity. Thus the coarse holes are exactly H .

For a canonical wall put $L = D \triangle \{j\}$ and $C = A \triangle L$. Formula (8) forces $z_A = z_C$ whenever both roles are nonempty proper. I give the finite-field obstruction explicitly. On a positive face, A contains j . For nonempty proper L , the cases $L = \{j\}$ and $L = I \setminus \{j\}$ already fail with $A = \{j, x\}$. In the remaining cases $A = \{j\}$ yields $\sum L = |L|j$. If $j \in L$, take $x \notin L$ and $A = \{j, x\}$; equality of means gives $(|L| - 2)(j - x) = 0$. Hence $|L| = 2$, and the previous sum forces the other element of L to equal j , impossible. If $j \notin L$, take $x \in L$; the equation is $(|L| + 2)(j - x) = 0$. Thus $|L| = p - 2$, and writing $I \setminus L = \{j, y\}$ in the first sum gives $y = j$, again impossible. Consequently L is empty or full.

For negative walls with $j \notin D$, inversion reduces to a positive wall. If $j \in D$, then $j \notin L$. For nonempty L , take $A = \emptyset$ and $C = L$. Formula (7) gives the child intercept $-\mu ag^{|L|} z_L$. A legal zero-translation wall has zero intercept, and its other factors are nonzero, so $z_L = 0$. Thus L is balanced. If $|L| \geq 2$, choose nonzero $x \in L$ and use $A = \{x\}$, $C = L \setminus \{x\}$; equality requires $x = -x/(|L| - 1)$, impossible. If $L = \{0\}$, choose $x \notin \{0, j\}$, giving distinct means for $\{x\}$ and $\{0, x\}$. Thus $L = \emptyset$, and $A = C = \emptyset$ forces the parent intercept to vanish.

The remaining possibilities are exactly (i)–(iii). They are sufficient: $C = A$ or $C = I \setminus A$ makes all applicable means agree, with the stated restriction for the empty role. All outer–outer fine contacts are legal zero-translation walls; central children are away from the box interface. Hole sufficiency follows from closure of E . Counting the disjoint lists gives (10). $\square$

Lemma 5.3 (Legality in complete tilings). *Every full $\text{CL}(E)$ -legal framed tiling is E -legal.*

Proof. Its hole catalog is still H . For a positive singleton wall T, U , the omitted cells have the same possible owner box. Full coverage supplies both owners; two different chairs cannot occupy that box, so $O(T) = O(U)$. Hence T and U are canonical outer siblings of one star and their contact is in the genuine P family.

For a positive complement wall on axis j , the actual owner $C = O(T)$ intersects U 's box in the cell $\mathbf{1} + e_j$, which U occupies. Therefore C must omit that cell, giving a second hole-owner edge $C \rightarrow U$. The balanced hole catalog forces $j = 0$ and the first owner's permutation to be $i + b$; the second edge has empty relative sign set and is H_0 . The original wall is exactly $B(b)$. Reverse walls give B^{-1} . Negative singleton walls and all holes were already in E . $\square$

The full-coverage qualification matters: a first corona need not include the hole-owner of its neighbor, and a locally compatible contact can fail to extend to a full tiling.

Proof of Theorem 1.1. Proposition 4.2 gives a unique first-level partition. Its parents are child-legal. Lemma 5.1 and connectedness of the cell-adjacency graph put all parent anchors in one common parity coset. Choose a reference origin and divide by two. The resulting tiling is $\text{CL}(E)$ -legal, hence E -legal by Lemma 5.3. Induction gives the unique hierarchy at every level.

For existence, set $b_j = 0$ for even j and $b_j = 1$ for odd j , and define

$$a_n = \sum_{j=0}^{n-1} 2^j b_j, \quad G_n = M_\mu^{-n}, \quad t_n = -a_n \mathbf{1}.$$

Take the scale- 2^n chair with frame G_n and anchor t_n , subdivided n times. In the next ancestor choose the empty role when $b_n = 0$ and the central role when $b_n = 1$. Its frame is $G_{n+1} M_\mu = G_n$ and its anchor is

$$-a_{n+1} \mathbf{1} + 2^n b_n \mathbf{1} = -a_n \mathbf{1}.$$

Thus these patches nest exactly. Their boxes range from $-a_n \mathbf{1}$ to $(2^{n+1} - a_n) \mathbf{1}$, and their omitted corners begin at $(2^n - a_n) \mathbf{1}$. Both a_n and $2^n - a_n$ tend to infinity; for $n = 2m$, $a_n = 2(4^m - 1)/3$. Every bounded box is eventually covered by occupied support. The union is a full E -legal tiling.

A translation preserving a framed tiling preserves its uniquely recognized parent partitions. At level n its vector lies in $2^n \mathbb{Z}^p$, since the parent anchors lie in one coset of this lattice. Therefore it is zero. Finally a frame-preserving Euclidean symmetry has linear part in the finite signed-permutation group. The homomorphism to linear parts has only translations in its kernel, now trivial, so the full framed symmetry group is finite. $\square$

6 A dense rational realization

The body T_p^{dense} uses the full signed-permutation orbit of tangent sites on every exposed facet. Theorem 1.2 concerns this specific decoration.

6.1 The full facet graph and the exact body

The construction first applies to any dimension $d \geq 3$ and finite proper registered contact language $\mathcal{A}$. There are $d2^d$ exposed unit facets of P_d . For a facet f , let b_f be its centre, n_f its outward normal, and I_f the matrix of the increasing tangent coordinate axes. Put $m = d - 1$ and

$$q_0 = \frac{(1, \dots, m)}{4d}, \quad Q = B_m q_0,$$

where B_m is the full signed-permutation group. Since the coordinates are distinct and nonzero, $q = A_q q_0$ determines a unique $A_q \in B_m$.

A site is a pair $u = (f, q)$. Across every shared unit facet of every contact in $\mathcal{A}$, join the transported sites by an edge. Each edge requires opposite signed heights. Set

$$\chi_u = \det[I_f A_q, n_f].$$

Proper contact transport reverses the outward normal and preserves the transported tangent frame, so χ changes sign across every edge. The graph is bipartite and has no forced-zero component. Order sites lexicographically by (b_f, n_f, q) ; order graph components by their least site, number them $1, \dots, J$, and set

$$a_u = j(u) \chi_u. \tag{11}$$

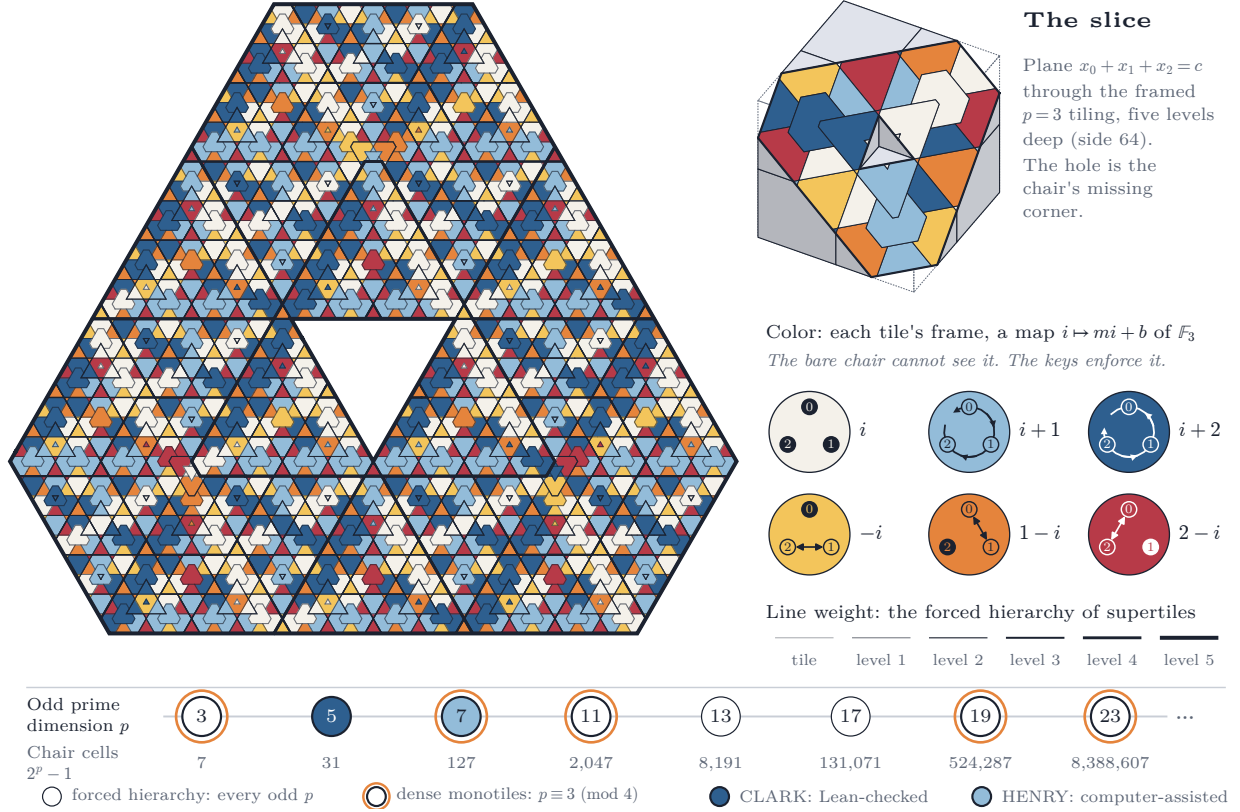


Figure 2: A planar section of a finite framed-carrier patch for $p = 3$, $\mu = 1$, $g = 2$. The plane $x_0 + x_1 + x_2 = 221/2$ cuts the chair $2^5 P_3$, of side 64, after five substitution levels; the triangular hole is its missing corner. Each coloured component belongs to one tile's carrier section. Colour records the affine index permutation $i \mapsto mi + b$ on $\mathbb{F}_3$; coordinate signs and physical boundary keys are suppressed. Line weights mark the nested supertiles. The hierarchy forced in every full registered tiling excludes nonzero translation periods, as proved in Theorem 1.1. The bottom strip summarizes the odd-prime dimensions covered by Theorems 1.1–1.3; the dense construction and HENRY are distinct bodies.

Different components have different absolute coefficients.

The shape can now be specified with rational numbers alone. Take the first $2m$ primes greater than J , in order $p_1^-, p_1^+, \dots, p_m^-, p_m^+$. Let N be the first prime greater than $32dJp_m^+$. Set

$$\eta = \frac{1}{128d}, \quad \rho = \frac{1}{32d}, \quad \delta = \frac{1}{16d}, \quad K = \prod_{i=1}^m [-\eta/p_i^-, \eta/p_i^+],$$

and

$$\psi(u) = \max\left(0, \min_i\{1 + p_i^- u_i/\eta, 1 - p_i^+ u_i/\eta\}\right).$$

At the site centre $c_u = b_f + I_f q$, replace the flat facet by the normal graph

$$z = \frac{\eta a_u}{N} \psi(A_q^{-1} I_f^{-1}(x - c_u)), \quad (12)$$

retaining material on its inward side. The inverse I_f^{-1} is used only on the tangent hyperplane. Outside these supports retain the carrier. This defines $T(\mathcal{A})$.

The supports are disjoint: distinct orbit centres differ by at least $1/(4d)$ in some coordinate, their half-widths are below η , and all centres are more than $1/4$ from the facet's integer edges. Different normal-axis tubes cannot meet because one would require a coordinate near an integer and the other requires it more than $1/4 - \eta$ away. Height is below ρ . In each support tube the map

$$(u, z) \mapsto (u, z + \max(0, 1 - |z|/\rho)h(u))$$

is an increasing homeomorphism in z and is the identity on the tube boundary. These maps combine to an ambient homeomorphism of the carrier to the keyed body. The carrier is a ball: it is strictly star-shaped from $(1/2, \dots, 1/2)$, and radial parametrization gives a homeomorphism with a closed ball. Thus $T(\mathcal{A})$ is a rational polyhedral d -ball. Each occupied cell retains its core $c + [\delta, 1 - \delta]^d$ in the interior.

6.2 Why the dense body registers arbitrary placements

Proposition 6.1 (Dense registration and exact correspondence). *Any tiling by arbitrary isometric copies of $T(\mathcal{A})$ becomes a proper signed-permutation, integer-translation carrier tiling after one ambient isometry. Its face contacts are exactly subject to the full site equations from (11). Conversely every registered carrier tiling satisfying these equations realizes a physical tiling by $T(\mathcal{A})$.*

Proof. The first step is to recover relative coordinate frames from coincident boundary features.

First, the slope bound is $L = Jp_m^+/N < 1/(32d)$. At a key's base crease, the absolute deviation from a flat material angle is $\alpha_{j,p} = \arctan(jp/N)$. At a side-side crease it is

$$\beta_{b,c} = \arccos \frac{1}{\sqrt{(1 + b^2/N^2)(1 + c^2/N^2)}}.$$

All deviations lie in $(0, \pi/4)$. Base deviations distinguish the pair (j, p) : if $jp = kq$ for two of the chosen primes, their being greater than J forces equality of both factors. A base deviation cannot equal a side-side deviation, since equality would give

$$N^2(a^2 - b^2 - c^2) = b^2 c^2,$$

where $0 < b, c < N$, contradicting primality of N .

Each tile has an interior ball of fixed radius, so volume packing proves local finiteness. At a generic point of a key crease remove the finitely many intersections with lower-dimensional strata of nearby tiles. A perpendicular two-plane then has an exact sector decomposition. Ordinary carrier sectors contribute multiples of $\pi/2$; a key sector lies strictly between $3\pi/4$ and $5\pi/4$. One key sector plus ordinary sectors cannot sum to 2π , three key sectors cannot fit, and two key sectors plus an ordinary one cannot fit. Thus exactly one other key sector is complementary. At a base crease the angle separation identifies the same magnitude and the same labeled base direction.

There can be no switch between different companion tiles at a lower-dimensional key junction. At any point of a key graph the tangent cone contains $\{(v, w) : w < -L\|v\|\}$. Its spherical fraction is at least

$$\frac{1}{2} - \frac{L\sqrt{d-1}}{\sqrt{2\pi}} > \frac{1}{3}.$$

Indeed use an isotropic Gaussian (V, W) , condition on $\|V\|$, bound the normal density by $1/\sqrt{2\pi}$ and use $\mathbb{E}\|V\| \leq \sqrt{d-1}$. Three such interior cones cannot be disjoint. The closed crease complex of one key is connected. Its intersections with the finitely many other tiles' key-crease complexes give a disjoint finite closed cover, by the generic complementary-sector result and closure. Consequently one companion key contains the whole crease complex.

The magnitude already identified makes the two complexes congruent. A compact metric space cannot contain a proper isometric copy of itself: if a point has positive distance from the image, its successive images are pairwise separated by that distance. Hence the complexes are equal. Their convex hulls are equal pyramids. The base has 2^m vertices, each side has $2^{m-1} + 1$, so the base and apex are intrinsic. The m base edge lengths are distinct; the apex projection differs from the box centre in every direction. Therefore the whole tangent frame is determined, with no remaining sign reversal.

Let (G, t) be the relative isometry across matched sites (f, q) and (f', q') . Recognition gives

$$GI_{f'}A_{q'} = I_fA_q, \quad Gn_{f'} = -n_f.$$

Thus G is a signed permutation. Site-centre equality cancels the common q_0 offset and gives $Gb_{f'} + t = b_f$. Facet centres are integral on their normal axis and half-integral on tangent axes, so t is integral. Opposite signed coefficients have equal magnitude and thus the same component label; taking determinants in the frame identity gives $\det G = 1$.

To pass from local companions to the entire tiling, start at one tile and follow key companions. Their carriers cannot share a cell because its retained core would lie in two interiors. The set of cells they own is nonempty and closed under every lattice step: across each exposed facet a matched key supplies the adjacent cell's owner. Thus it is all of $\mathbb{Z}^d$. Any additional tile contains a radius-1/4 interior ball centred at x . Clamp x into the retained core of the already registered cell containing it. The displacement is at most $\sqrt{d}\delta < 1/4$, producing an interior intersection. There is no additional tile.

Finally, in a registered tiling the two carrier owners across a facet must have opposite coefficients at every paired site, by coverage and disjoint interiors in the disjoint support tubes. Conversely these equations identify complementary normal graphs in every tube; outside the tubes the carriers already tile space. This proves the exact correspondence. $\square$

6.3 Exact facet closure when the dimension is three modulo four

Now take $d = p \equiv 3 \pmod{4}$, $\mu = 1$, $g = -1$, and $\mathcal{A} = E$. Write $F(E)$ for the proper, disjoint registered face poses satisfying every site equation on every shared unit facet. Here $\ell = 2$. Put

$\varepsilon_A = (-1)^{|A|}$ and $f_A(i) = \varepsilon_A(i - z_A)$, with the separately defined empty role. For these parameters, write $P(j, b)$ for a contact in $P_+ \cup P_-$ and $M(j) = M(-1, j)$.

The *tangent return group* at a facet is the group of tangent transformations induced by closed graph paths based at that facet.

Lemma 6.2 (Facet graph reduction). *The facet graph, before the free tangent sites are included, is a union of a component at inner label zero and components joining inner labels $k, -k$. The zero component has tangent return group of order two; each nonzero pair has trivial return group. The odd transport from inner label $-k$ to k is*

$$L_k = D_{\{k\}}P_-, \quad P_-(i) = -i,$$

with $L_{-k}L_k = \text{id}$.

Proof. Write V_k for the inner facet with centre $b_k = \frac{3}{2}\mathbf{1} - \frac{1}{2}e_k$, and $O(A, j)$ for the outer facet of cell A on axis j . The incoming hole contact

$$T_A = (h_A, 4A - \mathbf{1})$$

identifies $V_{f_A(j)}$ with $O(A, j)$. These edges form p stars, with every outer facet a unique leaf. Transport each leaf's ambient frame by h_A back to its inner label.

For a wall contact G from a companion outer facet $O(B, j')$ to $O(A, j)$, compute the transported matrix $h_A^{-1}Gh_B$. A $P(j, b)$ wall has $B = -A + b$, so $|B| = |A|$ and $z_B = -z_A + b$; the matrix has slope -1 , zero intercept and only axis $k = f_A(j)$ negative. An $M(j)$ wall gives the same matrix, including at $A = \emptyset$, which supplies all bridges $k \leftrightarrow -k$. A $B(b)$ wall has $B = (I \setminus A) + b$, opposite cardinality parity and $z_B = z_A + b$; the same multiplication again gives L_k . Inverses give L_{-k} .

These are all wall types. Successive nonzero bridges cancel. At zero, L_0 has order two and nontrivial tangent action $i \mapsto -i$ on nonzero labels. This determines the graph and every return transport. $\square$

Proposition 6.3 (Exact full-profile equality). *For this dense free-orbit construction, $F(E) = \text{CL}(E)$ as sets of poses.*

Proof. Opposite coefficients require an odd graph path. Since the tangent orbit is free, transporting a site fixes the entire tangent matrix; reversing the normal then fixes the whole proper ambient frame. Lemma 6.2 leaves exactly L_k as the odd inner transport.

An inner-to-outer mate is T_A or its inverse. An inner-to-inner mate would have $G = L_k, t = 2e_k$, and the same box as the root, so the two carriers overlap and it is rejected. For an outer-to-outer mate, the necessary equations are

$$G = h_A L_k h_B^{-1}, \quad f_B(j') = -k, \quad t = (4A - \mathbf{1}) - G(4B - \mathbf{1}) + 2h_A e_k. \quad (13)$$

Writing D for the negative set of G , one obtains

$$GB = A - D + h_A e_k, \quad t = 2D - 2h_A e_k.$$

Thus its lower box corner is $2(2A_j - 1)e_j$: every non-hole mate is a canonical wall.

Let $f_G(i) = ai + b$ and $K = D \triangle \{j\}$. For a wall side with fixed membership of j , its actual shared facets are exactly those proper A with $C = A \triangle K$ proper and $B = f_G(C)$. This follows by equating facet centres and opposing normals; the only absent panels are those where A or C is the full set. When A, C are nonempty, (13) gives $z_A = z_C$.

For completeness, the mean condition on every available facet forces $K = \emptyset$ or $K = I$. First suppose $j \notin K$ and $1 \leq m = |K| < p$. On the positive face, if $m = p - 1$, take $A = \{j, k\}$; its toggle is $I \setminus \{k\}$ and the means differ. If $m = p - 2$, write $I \setminus K = \{j, r\}$ and take $A = \{j\}$, whose toggle has mean r . If $m \leq p - 3$, $A = \{j\}$ forces $\sum K = mj$, and $A = \{j, k\}$ with $k \in K$ gives $(m + 2)(k - j) = 0$, impossible. On the negative face, a one-element complement gives a contradiction with $A = \{k\} \subset K$; a two-element complement $\{j, r\}$ gives one with $A = \{r\}$. With at least three complement elements, take distinct $r, s \notin K \cup \{j\}$; the singleton equations demand $\sum K = mr = ms$. All chosen toggles are nonempty proper. If $j \in K$, replace K by its complement and use $z_C = z_{I \setminus C}$. This proves the claim on both sides.

For $K = \emptyset$, $D = \{j\}$ and $a = -1$. Positive walls allow every intercept. On the negative side the empty–empty facet exists and forces $b = 0$. For $K = I$, $D = I \setminus \{j\}$ and $a = 1$; the actual facets have nonempty complementary roles, so every intercept is allowed on either side. These are exactly the walls of Proposition 5.2. Conversely, each listed wall satisfies (13) on every shared facet. The bridges supplied by the negative singleton contacts $M(-1, j)$, using their shared outer facet with role $A = \emptyset$, supply the odd path for each $k = j$. They transport every site, so all full profiles match. The holes already belong to E . Thus $F(E) = \text{CL}(E)$ as sets of poses. $\square$

Proof of Theorem 1.2. Apply Proposition 6.1 and Proposition 6.3. Every physical tiling gives a $\text{CL}(E)$ -legal carrier tiling, which is E -legal by Lemma 5.3. The registered hierarchy and explicit exhausting construction therefore prove the forced hierarchy and the existence of a physical tiling.

To transfer the absence of translations to unmarked tiles, suppose the physical tile collection has period $v \neq 0$. Bounded tiles have no finite orbit under this translation. Choose a native pose once on each orbit and propagate it by nv . The registration proof applies to these arbitrarily chosen poses, producing a framed realization with the same nonzero period, after the global change of coordinates. This contradicts Theorem 1.1.

Finally, registered copies have only finitely many orientations, and the polyhedron has finitely many facet normals. The signed unit normals in the whole tiling form a finite spanning set. Every full Euclidean symmetry permutes it, so its linear part belongs to a finite group. The map to linear parts has trivial translational kernel. The full symmetry group is therefore finite. Infinitely many dimensions occur: if primes congruent to three modulo four formed a finite list of product Q , the number $4Q - 1$ would have such a prime divisor absent from that list. $\square$

7 CLARK and HENRY

CLARK is the exact sparse body T_5 ; HENRY is the seven-dimensional body T_7 . The complete Lean verification of HENRY remains unfinished. The dense construction uses all $2^{d-1}(d - 1)!$ tangent sites on every exposed unit facet. The bodies in this section retain only a selected small set of sites and use one absolute key height. Their registration and finite mating laws must therefore be established separately.

7.1 An exact coordinate definition

Let $d \in \{5, 7\}$, $k = d - 1$ and $D = 4(k + 1) = 4d$. For $i = 0, \dots, k - 1$, set

$$\bar{x}_i = \frac{i + 1}{D}, \quad w_i = \frac{1 + i/k}{4D}, \quad o_i = \frac{w_i}{i + 3}, \quad h = \frac{1}{48d}. \quad (14)$$

The reference key K_0 is the convex hull of $Q_0 \times \{0\}$ and (o, h) , where $Q_0 = \prod_i [-w_i, w_i]$. At an exposed unit facet f with centre b_f , normal n_f and increasing tangent inclusion I_f , a placement

has tangent signed permutation γ and sign $\sigma \in \{1, -1\}$. Its base is

$$b_f + I_f \gamma(\bar{x} + Q_0)$$

and its apex is $b_f + I_f \gamma(\bar{x} + o) + \sigma h n_f$. A positive placement is a bump and a negative one is a dent. The complete finite placement lists are the two JSON files in Appendix A; these lists specify every placement.

If K^+ and K^- are the unions of the closed positive and negative key solids, the exact body is

$$T_d = \overline{(P_d \cup K^+)} \setminus K^-. \quad (15)$$

The closure retains the dent's inward sloping walls and apex; the relative interior of its base opening is absent. Equivalently, on disjoint facet collars the boundary is the closed normal graph of the signed pyramid profile. This is the convention used in the Lean definition.

For T_5 , 128 facets have one key and 32 have four; 128 keys are bumps and 128 are dents. For T_7 , 768 facets have one key and 128 have two; 512 keys are bumps and 512 are dents. All keys in a fixed dimension are congruent as solid pyramids. The edge margins are respectively 89/320 and 181/672, both greater than 1/4.

Figure 3 displays the reference key of CLARK and one exact bump-socket match. The four distinct tangent widths identify the tangent axes; the nonzero apex offsets distinguish their directions. The keys are shallow compared with the unit carrier cubes, so the figure enlarges the normal direction by a stated factor. These local views complement the key-free sections in Figure 1.

7.2 Why an arbitrary tiling must register

We prove the registration assertion directly for the sparse bodies just specified. Throughout this subsection $d \in \{5, 7\}$, $k = d - 1$, $D = 4d$, and $h = 1/(48d)$. Write

$$C_d = \{0, 1\}^d \setminus \{\mathbf{1}\}, \quad P_d = \bigcup_{c \in C_d} (c + [0, 1]^d).$$

An exposed unit facet is a triple $f = (c, a, s)$, where $c \in C_d$, $a \in \{0, \dots, d - 1\}$, $s \in \{-1, 1\}$, and $c + s e_a \notin C_d$. Its outward normal, centre and increasing tangent inclusion are denoted by $n_f = s e_a$, $b_f = c + \frac{1}{2} \mathbf{1} + \frac{s}{2} e_a$, and $I_f : \mathbb{R}^k \rightarrow e_a^\perp$, respectively. For each facet let $v_f : B_k \rightarrow \{-1, 0, 1\}$ be its specified sparse signed placement function: $v_f(\gamma) = 1$ records a bump, -1 a dent, and 0 an absent key. Here B_k is the group of signed coordinate permutations of $\mathbb{R}^k$.

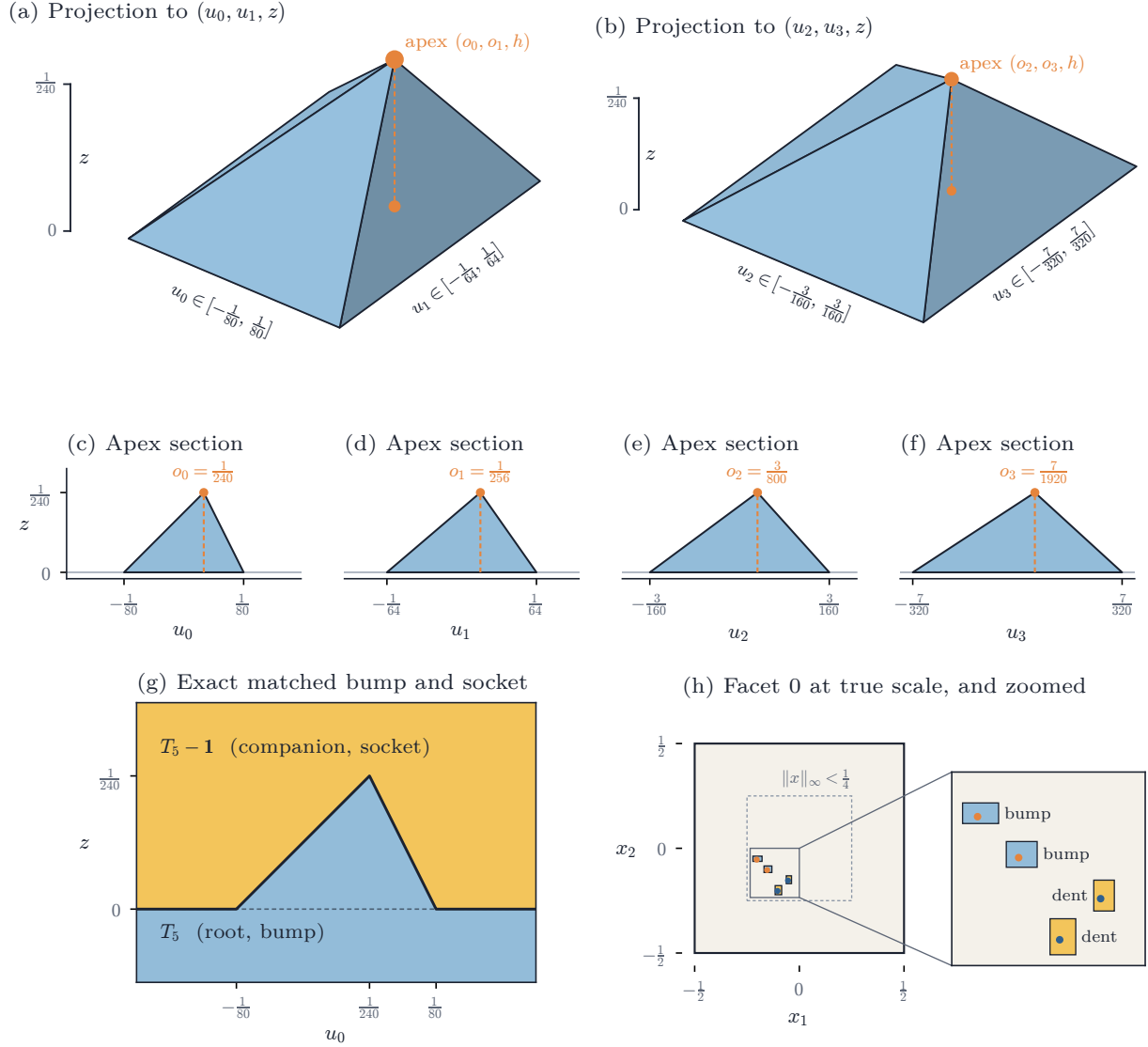
The following profile makes the local meaning of the closed-body convention explicit. For $u \in \mathbb{R}^k$, put

$$\kappa(u) = \max \left\{ 0, \min_{0 \leq i < k} \left\{ \frac{w_i - u_i}{w_i - o_i}, \frac{w_i + u_i}{w_i + o_i} \right\} \right\}, \quad H_f(x) = h \sum_{\gamma \in B_k} v_f(\gamma) \kappa(\gamma^{-1} x - \bar{x}).$$

The graph of κ above Q_0 is the lateral boundary of the pyramid with base Q_0 and apex $(o, 1)$. In the facet collar

$$X_f = \{b_f + I_f x + \tau n_f : \|x\|_\infty \leq \frac{1}{4}, |\tau| \leq \frac{1}{8}\},$$

the body is the closed subgraph $\tau \leq H_f(x)$; outside the collars it agrees with P_d . In particular a dent is an actual opening into the carrier. Its open base is not an additional boundary face.



Panels (a)–(g): normal height enlarged $4 \times$ relative to tangent lengths; labels give unscaled coordinates.

Figure 3: CLARK's reference key and a matching socket. (a),(b) are coordinate projections of K_0 ; each also equals the section fixing the omitted tangent coordinates at $u_j = o_j$. (c)–(f) retain (u_i, z) with all other $u_j = o_j$, on a common scale. (g) shows the local material of T_5 (blue) and $T_5 - \mathbf{1}$ (gold) on opposite sides of their common profile; the matched keys have all 17 vertices coincident. The dashed line $z = 0$ is the carrier plane, a reference line, not a retained socket surface. (h) shows facet 0 of T_5 (cell $\mathbf{0}$, outward normal $-e_0$) at true scale and in a magnified inset, projected to the centred tangent coordinates along world axes (x_1, x_2) : the displayed coordinates subtract $1/2$ from the corresponding world coordinates. Its four key bases comprise two bumps and two dents, with their apex points marked. The dashed square indicates the centred quarter-box that contains every key support. In (a)–(g), normal height is enlarged fourfold relative to tangent lengths; tick labels give unscaled coordinates.

Lemma 7.1 (The shallow-key estimates). *For each of the two specified bodies the following assertions hold.*

1. *Every exposed unit facet has a key. Distinct keys on a facet have disjoint closed supports, each at distance greater than $1/4$ from the edge of that facet. Collars of distinct exposed facets are disjoint.*
2. *A key solid has diameter less than $1/4$.*
3. *Each occupied unit cell retains the open core $c + (h, 1-h)^d$. The open ball of radius $r = 1/2 - h$ about its centre is contained in the interior of the body. Moreover $2dh < r$.*
4. *Every key-side slope is at most $1/2$. At a genuine key crease the material angle is $\pi + \delta$ or $\pi - \delta$, where $0 < \delta < \pi/4$. At a genuine carrier ridge the material angle is $\pi/2$ or $3\pi/2$.*

Proof. The assertion that no exposed facet is empty is the following small finite part of the coordinate specification. Form C_d by its binary definition, form all triples (c, a, s) with $c + se_a \notin C_d$, and compare these triples with the facet rows of the placement table. There are $d2^d$ of them. For $d = 5$ every one of the 160 rows has one or four keys, with respective multiplicities 128 and 32; for $d = 7$ every one of 896 rows has one or two, with multiplicities 768 and 128. The comparison uses the triples, not merely the number of rows. In each listed key, the k tangent indices are a permutation of $0, \dots, k-1$, every tangent sign is ± 1 , and the signed height is $\pm h$. These are finite assertions about the defining tables; no assertion about tilings enters their verification.

All remaining bounds follow from the formula for the reference key. The largest possible tangent coordinate of a support is

$$M = \frac{k}{D} + \frac{2 - 1/k}{4D} < \frac{1}{4}.$$

Consequently its edge margin is $1/2 - M > 1/4$. These margins are $89/320$ and $181/672$ for $d = 5$ and 7 . The vectors $\gamma \bar{x}$, $\gamma \in B_k$, are distinct, and different such vectors differ by at least $1/D$ in some coordinate. On the other hand $2 \max_i w_i < 1/D$. Hence even the full signed orbit of supports is pairwise disjoint; its sparse subset is disjoint as well.

For two coplanar collars, their tangent centres differ by an integer in at least one coordinate, whereas their half-widths are $1/4$. Parallel distinct carrier planes are at least one unit apart. Finally, for perpendicular collars, a coordinate normal to one is a tangent coordinate of the other: in the first collar it is within $1/8$ of an integer, and in the second its distance from the integers is at least $1/4$. This proves collar disjointness.

A base-to-base distance squared is at most $4 \sum_i w_i^2$. An apex-to-base-vertex distance squared is at most $\sum_i (w_i + o_i)^2 + h^2 < 4 \sum_i w_i^2 + h^2$. The diameter of a convex hull is the maximum of its vertex distances, so

$$\text{diam}(K_0)^2 < \frac{k}{D^2} + h^2 < \frac{1}{16} \quad (d = 5, 7).$$

A dent penetrates at most h from its carrier plane and a bump removes nothing. This proves the core assertion. A point at Euclidean distance less than $1/2 - h$ from a cell centre has every coordinate strictly between the corresponding core endpoints. Also $2dh = 1/24 < 1/2 - h$.

The minimum of $w_i - o_i$ is attained at $i = 0$ and equals $1/(24d)$; both factors in $w_i - o_i = (1 + i/k)(i + 2)/(16d(i + 3))$ increase with i . The slopes $h/(w_i \mp o_i)$ are therefore at most $1/2$. At a key base the angle deviation is $\alpha = \arctan s$ for one such slope s . At a ridge between side faces from two different tangent axes it is β , where

$$\cos \beta = \frac{1}{\sqrt{(1 + s_1^2)(1 + s_2^2)}}.$$

Thus $0 < \alpha < \pi/4$ and $\cos^2 \beta \geq 16/25 > 1/2$, giving $0 < \beta < \pi/4$. A bump has base angle $\pi + \alpha$ and side-side angle $\pi - \beta$; a dent reverses these signs. Opposite side faces for one tangent axis meet only at the apex. The remaining genuine ridges are those of the orthogonal carrier, with angles $\pi/2$ or $3\pi/2$. The edge margin prevents key creases from meeting carrier ridges. $\square$

Fix a tiling $\mathcal{T}$ by the exact body, and choose, arbitrarily, an isometry g_A carrying the native body onto each $A \in \mathcal{T}$. At this stage the isometries may have any rotations, reflections and translations. No face-to-face property is assumed. A *key side* means one of the $2k$ closed lateral faces of a key pyramid; a *key crease* is a codimension-one face within such a side. Carrier boundary patches are allowed to be nonconvex and perforated. Only their local planar pieces, not an unjustified convex-facet structure, will be used.

Lemma 7.2 (Local finiteness and transverse sectors). *The tiling is locally finite. On the relative interior of a key side F , of supporting plane Π , there is a connected dense set on which only the following possible obstructions to a flat companion remain: ridges of other tiles whose affine spans lie in Π . At a generic point of a key crease, a perpendicular two-plane cuts the incident tiles into sectors with disjoint interiors, covering the plane. Each sector angle belongs to*

$$\Theta = \{\pi/2, \pi, 3\pi/2\} \cup \{\pi - \delta, \pi + \delta : 0 < \delta < \pi/4 \text{ is a key deviation}\}.$$

In particular every sector angle is at least $\pi/2$.

Proof. Each tile contains an open ball of the fixed radius $r > 0$ from Lemma 7.1. If a tile meets a bounded set, this ball lies in a fixed bounded enlargement, because all tiles have the same finite diameter. The balls have disjoint interiors. A volume bound on finite subfamilies therefore bounds the number of tiles meeting the set, proving local finiteness.

Only finitely many tiles meet a neighbourhood of F . Inside Π remove their faces of dimension at most $d - 3$, their intersections with ridge spans not contained in Π , and the intersections of any two distinct ridge spans contained in Π . Enlarge each removed piece to an affine subspace of Π if necessary. All these subspaces have dimension at most $d - 3$, hence codimension at least two in Π . Their removal leaves a dense connected subset of relint F . Indeed, given two surviving points x, y , choose $z \in \text{relint } F$ outside the finitely many affine spans of x or y with a removed subspace. Each span is proper. The segments $[x, z]$ and $[z, y]$ stay in relint F and avoid the removed sets.

For a crease of affine span Λ , remove instead its intersections with lower-dimensional faces, ridge spans different from Λ , and facet hyperplanes not containing Λ . These form a finite union of proper affine subspaces of Λ ; the complement is dense. Connectedness is not required here. At any surviving point every incident tile is locally either a half-space or a ridge wedge along Λ . It cannot contain the point in its interior, since the original tile has interior points arbitrarily close. Nonincident tiles stay a positive distance away. Thus a sufficiently small perpendicular disk is exactly partitioned by the asserted sectors. The angle inventory follows from Lemma 7.1; artificial subdivisions of a flat patch merely give angle π . $\square$

Lemma 7.3 (A key side cannot be divided between companions). *For every key side F of a tile A there is a single tile $B_F \neq A$ on its other side. It contains all of F , and F is contained in one of its maximal flat boundary patches.*

Proof. Let ν point out of A across F . At a generic point of F , A occupies the negative half-space of Π . A facet of another tile through that point must lie in Π ; otherwise its material half-space would intersect the interior of A . If no other tile has a facet there, the generic-point construction puts all incident ridge spans in one common $(d - 2)$ -plane $\Lambda \subset \Pi$. In a perpendicular section, their

angles must sum to π . From Θ there are only two possibilities: a single angle π , or two angles $\pi/2$. Indeed every angle less than π other than $\pi/2$ is strictly greater than $3\pi/4$.

We now exclude the second possibility; this is the step that prevents a key side from being split between two carrier edges. One of the two putative tiles, C , has a carrier face in Π adjoining its right-angle ridge. Choose $x' \in \Lambda$ close to the original point, still in relint F , and in the relative interior of one unit subdivision of that carrier ridge. Let U be a small convex relatively open neighbourhood of x' in that subdivision, and let $u \in \Pi$ be the inward coordinate direction along the adjacent exposed carrier face of C , perpendicular to Λ . The strip

$$\Omega = \{z + tu : z \in U, 0 < t < 1/4\}$$

is relatively open in Π . It is untouched by keys of C , since all their supports are more than $1/4$ from an edge. Hence C is locally exactly the positive- ν half-space along Ω .

The strip meets relint F for small t . It also has points outside F : choose t larger than $\text{diam } F$ and still below $1/4$. Join an interior strip point to an exterior one by a path in the strip avoiding the codimension-two boundary skeleton of the convex polytope F . Such a path exists by the same intermediate-point argument used above. Its first exit crosses the relative interior of a boundary face e of F , hence a genuine key crease of A . Since Ω is open, we can choose the crossing point generic on e .

If the angle of A there is $\pi + \delta$, its material crosses into the half-space occupied by C . If its angle is $\pi - \delta$, the two tiles leave a sector of angle $\delta < \pi/2$. No remaining sector can fill it, since every possible sector has angle at least $\pi/2$. Both cases contradict the tiling.

Every generic point of F therefore has exactly one flat companion. Its identity is locally constant there: by local finiteness, a small neighbourhood meets no tile except those already incident, and the companion occupies the opposite open half-space. Connectedness makes the identity independent of the point. Closure gives $F \subset B_F$. Every point of F is a boundary point of B_F , since interior points of A accumulate at it. The locally flat pieces form one connected planar sheet. If their plane is a carrier plane of B_F , they belong to its flat carrier boundary; otherwise the sheet belongs to a single key side, because the different keys of B_F are positively separated. This proves the asserted containment; equality of faces has not been assumed. $\square$

Lemma 7.4 (There is only one complementary tile at a key crease). *At a generic point of a key crease of A , precisely one other tile occupies the complementary sector. In particular the companions of two incident key sides are the same tile.*

Proof. If A has angle $\pi + \delta$, the complement has angle $\pi - \delta < \pi$; two sectors, each at least $\pi/2$, cannot fit. If A has angle $\pi - \delta$, the complement has angle $\pi + \delta < 5\pi/4$, so at most two sectors fit. A two-sector partition could only use $\pi/2$ and $\pi - \delta'$, or $\pi - \delta_1$ and $\pi - \delta_2$. The resulting equations are

$$\delta + \delta' = \pi/2, \quad \text{or} \quad \delta + \delta_1 + \delta_2 = \pi,$$

respectively. They are impossible because all deviations are strictly less than $\pi/4$. The combinations involving a flat or reentrant sector are at least $3\pi/2$, and two right angles total only π . Thus the complementary sector is unique and has a genuine ridge. The companion supplied by Lemma 7.3 contains the whole closed side and hence its crease. It must be this unique complementary tile. Apply this observation to both incident key sides. $\square$

Lemma 7.5 (Recognition of an entire key). *Every key of every tile has one companion key whose closed solid pyramid coincides with it. The two keys have opposite types, bump and dent.*

Proof. Label the $2k$ sides by $(i, +)$ and $(i, -)$. Sides with different indices i meet along a side-side crease. This adjacency graph is connected for $k \geq 2$, so Lemma 7.4 gives a single tile B opposite every side of the chosen key of A . Let p be its apex.

We first prove that none of the companion flat patches is carrier boundary of B . Suppose the patch opposite a side F were in a carrier plane Π of B . At a generic point of any side-side crease e of F , the unique complementary B ridge has angle $\pi \pm \delta$, not $\pi/2$ or $3\pi/2$. Since it adjoins flat carrier boundary, it must be a base crease of a key of B .

This is initially a generic local assertion, and it must be extended before making any claim at the apex. The finite union of the closed base-crease sets of B contains a dense subset of the closed crease e , hence all of e . Distinct keys of B are disjoint compact sets and therefore positively separated. The connected set e must lie in a single such key. Its closure contains p . All side-side creases of F consequently belong to the same key of B , since two different keys cannot both contain p . Let $Q \subset \Pi$ be this key's base box.

In the plane Π take the tangent cones K of F and L of Q at p . Near p , every boundary face of F is a side-side crease: the base of F stays a positive distance from p . The preceding paragraph gives $\partial K \subseteq \partial L$. Moreover $\text{int}_\Pi K$ and $\text{int}_\Pi L$ are disjoint. Otherwise a sufficiently short common interior ray would lie both in the flat carrier boundary of B and in the open base of its key. The latter is absent from that flat boundary for both a bump and a dent.

This is impossible by convexity. The cone K is the pointed cone over the bounded base of the side-face pyramid F . Choose a nonzero vector v parallel to that base. Such a vector exists because $\dim F = k \geq 2$. For any $z \in \text{int}_\Pi K$, the line $z + \mathbb{R}v$ meets K in a bounded nondegenerate interval: the height above the apex is constant along the line, and the section of K at that height is a bounded homothetic copy of the base of F . The two endpoints belong to $\partial K \subseteq L$. Convexity of L puts z in L . Since this holds for the open set $\text{int}_\Pi K$, that set lies in $\text{int}_\Pi L$, a contradiction.

Every companion patch is therefore a side face of a key of B . All these closed faces contain p , so positive separation again puts them on one key. The $2k$ side hyperplanes of the key of A are distinct. Their companion faces must be distinct, and thus exhaust the $2k$ side faces of that one key of B . Their common intersection is the apex, so both apices are p .

Take now a generic point on a base crease of the key of A . The complementary ridge of B is incident to the already matched side plane, and its other plane is the base plane of the key of A . It cannot be a side-side ridge of the key of B : all its side planes have just been identified with those of the key of A , and none is that base plane. It is a base crease, so the two keys also have the same base plane. Intersecting the common side planes with this plane recovers the same bounded box. The common apex and base prove equality of the solid pyramids.

Along a side interior the material of a bump lies towards its solid pyramid, whereas the material of a dent lies away from it. The two tiles occupy opposite sides there. Thus one key is a bump and the other a dent. This also rules out a collection of partial contacts that covers only separate pieces of a key: a whole coincident companion has been deduced from those local contacts. $\square$

Lemma 7.6 (One key fixes the whole relative placement). *If a key of A on native facet f and with tangent frame γ has a companion in B on native facet f' with tangent frame γ' , then the relative isometry $g_A^{-1}g_B(x) = Gx + t$ satisfies*

$$G[I_{f'}\gamma', n_{f'}] = [I_f\gamma, -n_f], \quad Gb_{f'} + t = b_f. \quad (16)$$

In particular $G \in B_d$ and $t \in \mathbb{Z}^d$. The entire unit facet f' maps onto f , with opposed outward normals.

Proof. Whole-key coincidence maps the base box and its centre to the base box and its centre, and the apex to the apex. Orthogonal projection then maps the apex foot to the apex foot. The distinct

edge lengths $2w_i$ identify each tangent axis. Hence initially $GI_{f'}\gamma'e_i = \epsilon_i I_f \gamma e_i$ with $\epsilon_i \in \{-1, 1\}$. Comparing the apex-foot offsets from the base centres gives $\epsilon_i o_i = o_i$. Since every o_i is nonzero, every ϵ_i is $+1$. The signed normal heights have equal magnitude and opposite types, so $Gn_{f'} = -n_f$. These facts give the first identity in (16).

The base centres are $b_f + I_f \gamma \bar{x}$ and $b_{f'} + I_{f'} \gamma' \bar{x}$. Their equality after transport and the tangent-frame identity cancel the $\bar{x}$ terms, giving the second identity. Both displayed frames are signed coordinate bases, so G is a signed permutation. A facet centre has one integer coordinate, in its normal direction, and half-integer coordinates in every tangent direction. The map G takes normal direction to normal direction. Thus every coordinate of $b_f - Gb_{f'}$ is an integer. This proves the assertion about t .

A unit facet is its centre plus the tangent cube of half-width $1/2$, which is preserved by a tangent signed permutation. Consequently the whole facets coincide, and the carrier cell of B incident to f' is exactly the exterior unit cell across f . This eliminates continuous slides and arbitrary residual rotations; no preferred choice of the representatives g_A, g_B was made. $\square$

Lemma 7.7 (One neighbour and the complete profile on a unit facet). *Every exposed unit facet of A has one companion tile B owning the exterior unit carrier cell. All keys of the two facing facets match, including the absence of keys. If $m = I_{f'}^\top G^\top I_f$ is the tangent map from the root chart to the companion chart, then*

$$H_f(x) = -H_{f'}(mx) \quad (x \in [-1/2, 1/2]^k). \quad (17)$$

Proof. There is at least one key on f . For every such key, Lemma 7.6 puts its partner's incident carrier cell in the same exterior unit cell $c + se_a + [0, 1]^d$. Two different partner tiles would both contain that cell's open core in their interiors. Thus all keys have the same partner B . For this fixed relative placement the facet f' facing back towards f and the tangent map m are unique. Every occupied site of f therefore has its complementary key on f' .

Suppose, conversely, that f' had a key opposite an empty site of f . Apply Lemmas 7.5 and 7.6 to this key of B . Its partner owns the original carrier cell $c + [0, 1]^d$ of A . Their open cores force that partner to be A . Whole-key equality then gives the allegedly absent key on f , a contradiction. Thus the correspondence of supported sites is bijective and preserves the zero sites too. Equivalently,

$$v_f(\gamma) = -v_{f'}(m\gamma) \quad (\gamma \in B_k).$$

Substituting this identity into the definition of H_f proves (17). In a facet collar the two actual bodies are now the complementary closed subgraphs $\tau \leq H_f(x)$ and $\tau \geq H_f(x)$. They cover the collar with disjoint interiors. Full-facet matching is a conclusion of the proof, including at key boundaries and at zero sites. $\square$

Proposition 7.8 (Registration of the exact sparse bodies). *Let $\mathcal{T}$ be a tiling by T_d , $d \in \{5, 7\}$, and choose arbitrary native representatives $A = g_A(T_d)$. There is one ambient isometry e such that, for every tile A ,*

$$e \circ g_A(x) = G_A x + t_A, \quad G_A \in B_d, \quad t_A \in \mathbb{Z}^d.$$

The corresponding carriers partition the lattice unit cells. Every pair of carriers sharing a unit facet satisfies the complete profile identity (17). Conversely, any registered carrier tiling satisfying all these identities is a tiling by the exact decorated body.

Proof. Choose one tile A_0 and take $e = g_{A_0}^{-1}$. Let $\mathcal{C}$ be the tiles reached from A_0 by finitely many of the facet-companion steps of Lemma 7.7. Lemma 7.6 shows that every tile in $\mathcal{C}$ is registered in this one lattice. Two such carriers cannot own the same lattice cell, since its open core would belong to both tile interiors.

Let $W \subset \mathbb{Z}^d$ be their set of owned cell corners. It is nonempty. If $q \in W$, each cell at $q \pm e_i$ either belongs to the same carrier or lies across one of its exposed unit facets. In the latter case its owner is the facet companion, also in $\mathcal{C}$. Thus W is closed under every positive and negative coordinate step. The lattice graph is connected, so $W = \mathbb{Z}^d$.

It remains to exclude a separate collection of unregistered tiles; connectivity of a physical contact graph has not been assumed. Take any tile S and the centre x of one of its carried cells. The ball $B(x, r)$ lies in $\text{int } S$. Choose a lattice unit cell $q + [0, 1]^d$ containing x , and let $A \in \mathcal{C}$ be its owner. Clamp each coordinate of x into $[q_i + 2h, q_i + 1 - 2h]$, obtaining y . Then

$$y \in q + (h, 1 - h)^d \subset \text{int } A, \quad \|y - x\|_2 \leq 2\sqrt{d}h \leq 2dh < r.$$

Thus $y \in \text{int } S \cap \text{int } A$, forcing $S = A$. Every tile has been reached, and every originally chosen representative is registered. The unique owner of the exterior cell at each exposed facet is its already constructed companion, so all shared-facet identities hold.

For the converse, the global collars of a registered carrier partition are disjoint except that the two copies at a shared facet are the same collar. This follows from the coordinate separation argument in Lemma 7.1. On each such collar the opposed profiles agree, and the two closed subgraphs are complementary. Outside the collars the bodies agree with their carriers. They therefore cover space and have pairwise-disjoint interiors. Contacts of lower dimension require no additional compatibility: no key collar reaches a unit-facet edge, and closure supplies all remaining boundary points. $\square$

The proof applies to both determinant signs from its first step. It proves registration and complete physical profile enforcement, but it does not alone identify the resulting finite mating catalog. That identification is the separate finite argument below.

8 The five-dimensional proof

The geometric argument above converts any tiling by CLARK into a tiling by registered chairs whose contacts can be checked exactly. We now prove, directly in dimension five, that these contacts force parents of arbitrarily large size. The same contact relation also gives an explicit construction of a tiling.

8.1 The exact contact relation

A *relative pose* (G, t) specifies the second carrier as $GP_5 + t$ when the first is P_5 . Let M consist of those poses for which the carriers have disjoint interiors, meet along a four-dimensional face, and have complementary key profiles on every shared unit facet. The profile at an unoccupied key site is zero. Thus testing a contact means comparing the whole boundary profile, not finding just one bump that fits one socket.

The enumeration is finite for a geometric reason. Every actual contact has a matched key pair, and equation (16) determines its entire relative pose. We therefore enumerate all ordered pairs from the 256-key placement list, recover each pose, remove duplicates, and test carrier overlap and every shared profile. All coordinates and tests are rational. The registration proof establishes that this procedure omits no physical contact.

Finite computation 8.1 (The exact mating relation). For the CLARK coordinate list, M has 284 poses, all proper, and every relative translation has constant coordinate parity. The permutation

$R : i \mapsto -i$ of the five coordinate indices is a symmetry of the keyed body. With $K = \{\text{id}, R\}$,

$$M = K \text{CL}(E_5)K, \quad |\text{CL}(E_5)| = 222.$$

The same exhaustive procedure in dimension seven gives $M_7 = \text{CL}(E_7)$ with 408 poses, all proper. Here E_5 uses $(p, \mu, g) = (5, 1, 2)$ and E_7 uses $(7, 1, -1)$; their sizes are 116 and 318.

The five-dimensional proof below uses all 284 poses. Since R preserves the body, we identify representatives (G, t) and (GR, t) . These are right- K classes of chosen frames; no classification of all possible self-isometries of the body is needed. Membership in M is unchanged when either endpoint representative is multiplied on the right by R .

The exact enumeration has a second useful description. Normalize a root chair to P_5 , identify a binary cell vector with a subset $A \subseteq \mathbb{F}_5$, and let $O(T)$ denote the unique tile occupying a tile T 's omitted cell. The incoming contacts, namely tiles whose omitted cell is occupied by the root, have one possible right- K class for each proper A . The incoming tile for A has bounding-box lower corner $2\mathbf{1}_A - \mathbf{1}$ and omits cell A . Apart from these incoming contacts and their outgoing inverses, the remaining contacts lie in boxes with lower corner $2\varepsilon e_j$, where $\varepsilon = \pm 1$. Their four possible omitted cells are

$$\mathbf{1} + e_j, \quad \mathbf{1} - 3e_j, \quad 3e_j, \quad -e_j, \quad (18)$$

for positive singleton, negative singleton, positive complement and negative complement walls respectively. Here positive or negative refers to ε ; a singleton wall reverses only the contact-axis coordinate, while a complement wall reverses the other four coordinates. These statements are finite identities of the enumerated pose list. The reconstruction algorithm, candidate counts and exact certificates are given in Appendix C.

8.2 The 32-child dissection

Index the coordinates by $\mathbb{F}_5$. For a nonempty proper subset A , define its mean

$$z_A = |A|^{-1} \sum_{i \in A} i \quad \text{in } \mathbb{F}_5, \quad z_\emptyset = 0.$$

The mean is defined because $1 \leq |A| \leq 4$. For each of the 31 proper subsets, put

$$(h_A x)_i = (-1)^{\mathbf{1}_A(i)} x_{2|A|(i-z_A)}, \quad u_A = 4\mathbf{1}_A. \quad (19)$$

Add the central child $(h_c, u_c) = (\text{id}, \mathbf{1})$. In particular the empty outer child is $(\text{id}, 0)$.

All child frames are proper. Translation of the indices in $\mathbb{F}_5$ is an even permutation, while multiplication by 2 is a four-cycle on the nonzero indices and is odd. The permutation in h_A consequently has sign $(-1)^{|A|}$, which cancels the $|A|$ coordinate sign reversals.

These children dissect $2P_5$. The outer child with role A occupies the side-two box with cell set $2\mathbf{1}_A + \{0, 1\}^5$, except for the single cell $\mathbf{1} + \mathbf{1}_A$. The 31 outer boxes are disjoint and are exactly the macrocell boxes of $2P_5$. The central child occupies their 31 omitted cells. This proves the dissection, including disjointness of interiors. The parent represents the union of these 32 carrier copies; its frame and anchor are the data retained at the next level.

The dissection is well-defined on right- K classes. Since $z_{RA} = -z_A$,

$$Rh_A R = h_{RA}, \quad Ru_A = u_{RA}, \quad R\mathbf{1} = \mathbf{1}. \quad (20)$$

Changing the parent frame by R therefore permutes the children and changes their representatives only by right R .

For a tile C with representative (G, t) , let $S(C)$ be the proposed 32-child patch with C central. Its doubled parent has frame G and anchor $t - G\mathbf{1}$. The anchor is unchanged by replacing G by GR . The incoming part of the computed contact list consists exactly of the 31 outer children in this patch, read modulo K .

8.3 Every tile belongs to one complete parent

Proposition 8.2 (Recognition of parents). *Every full registered M -legal tiling, with its chosen frames read modulo K , has a unique partition into complete copies of the 32-child patch. The partition commutes with translations of the framed tiling.*

Proof. A tile T has only two possible parents: $S(T)$ and $S(O(T))$. Indeed, if T is not central, the omitted cell of T is occupied by the central tile of its parent. Full coverage determines that tile as $O(T)$, and its pose determines the entire proposed parent.

Normalize T to P_5 . Every contact in M has all translation coordinates of the same parity. The face-adjacency graph is connected: join occupied unit cells along the connected lattice cell graph; whenever two consecutive cells have different owners, those owners share a unit facet. It follows that all carrier boxes lie in the two parity classes 0 and $\mathbf{1}$ modulo 2. There is at most one tile in a given side-two box: two chairs in the same box would share at least 30 occupied cells.

Write I_A for the proposed incoming tile whose box has lower corner $2\mathbf{1}_A - \mathbf{1}$, and W_j^ε for the possible tile in the box with lower corner $2\varepsilon e_j$. For every proper A and axis j , consider the exterior cell

$$x(A, j) = \mathbf{1}_A + (2\mathbf{1}_A(j) - 1)e_j.$$

Checking its five coordinates against the two box parities leaves exactly two possible owner boxes: I_A and $W_j^{2\mathbf{1}_A(j)-1}$. A tile in the first box must omit A , already occupied by T , so it is precisely the catalogued incoming tile. Thus absence of I_A forces the indicated wall to be present.

Suppose a nonempty incoming tile I_A is present. It occupies every cell $x(A, j)$. For nonempty proper A , none of the four wall holes in (18) equals this cell. Hence all corresponding walls are absent. If another proper subset B agrees with A in any coordinate, its associated wall is absent too, forcing I_B . Only the complement of A agrees nowhere. Choose $j \in A$ and $k \notin A$; the proper nonempty subset $\{j, k\}$ agrees with both A and its complement, so it forces that last incoming tile as well. Therefore one nonempty incoming tile forces all of $S(T)$.

If $S(T)$ is incomplete, only $I_\emptyset$ can be present, and all ten walls $W_j^\pm$ are present. If $I_\emptyset$ is present, it occupies $-e_j$, so every negative wall must omit that cell and must be a negative complement wall.

Put $C = O(T)$. If T has a nonempty role in $S(C)$, the preceding argument completes $S(C)$. The remaining contact is the empty-role contact; choose representatives so that C has the same frame as T and translation $\mathbf{1}$. Then T is the empty incoming tile of C . If $S(C)$ were also incomplete, its negative walls would be complement walls. In T 's coordinates, the j -th such wall omits $\mathbf{1} - e_j$, an occupied cell of T . This is a nonempty incoming tile of T , contradicting incompleteness of $S(T)$. At least one candidate parent is complete.

They cannot both be complete. Normalize $S(C)$ to the standard doubled chair and let T have outer role A . The box of $S(T)$ has cell-coordinate intervals $[-1, 2]$ off A and $[1, 4]$ on A , with omitted block $\{1, 2\}^5$. Choose $j \notin A$ and $k \neq j$. Take $x_j = 0$, take $x_k = 2$ if $k \notin A$ and $x_k = 1$ if $k \in A$, and choose the other coordinates inside T 's box. This cell belongs to both parent supports but to neither T nor C . Its two proposed owners are noncentral children. The outer child boxes of $S(C)$ have even lower corners. The anchor of $S(T)$ is $6\mathbf{1}_A - \mathbf{1}$, so its outer child boxes have odd lower corners. They are distinct tiles, contradicting disjoint interiors.

Thus each tile belongs to exactly one complete parent. Two parents sharing a tile would be its two forbidden complete candidate parents, so the parents partition the tiling. The construction uses only actual omitted-cell owners and the chosen frames modulo K ; translating the tiling translates the resulting partition. $\square$

8.4 Coarsening preserves the same contact relation

Recognition alone gives only one level. To iterate it, we must prove that the parents have even relative offsets and that halving them gives another M -legal tiling.

Finite computation 8.3 (Refinement and coarsening). All sibling contacts in the 32-child patch belong to M . Among disjoint, face-touching canonical carrier parent patches whose cross-child contacts all belong to M , every relative translation is even. Dividing it by two gives exactly the relation M . Conversely, every contact in M refines into contacts in M .

Here is why the finite test is exhaustive, including alignment. Write a relative parent pose before halving as (F, v) , with $v \in \mathbb{Z}^5$; integrality follows from the central-child anchors. Some child r of one parent meets a child s of the other. If their relative contact is $(G, t) \in M$, direct composition gives

$$F = h_r G h_s^{-1}, \quad v = u_r + h_r t - F u_s.$$

The formula also proves alignment without assuming it. Every u_r is either even in all coordinates or equal to $\mathbf{1}$, and every translation in M has constant coordinate parity. Signed coordinate permutations preserve that property, so all coordinates of v have the same parity. If they were odd, the first parent's carrier boundary planes would have even coordinate offsets and the second parent's would have odd offsets. No facets could coincide, contradicting their positive four-dimensional contact. Hence v is even.

Every possible parent pair therefore occurs among the $32^2 \cdot 284$ choices of $(r, s, (G, t))$. For each distinct candidate, enumerate its occupied unit cells, reject an interior overlap, and compare the pose of every cross-parent child pair sharing a unit facet with M . This checks the whole common boundary. The 284 survivors all have even v , and $(F, v/2)$ runs through M exactly. No even-offset hypothesis is imposed on the candidates. The finite counts and checker specification are in Appendix C.

Apply this result to neighboring parents from Proposition 8.2. The parent face-adjacency graph is connected, so their anchors lie in one coset of $2\mathbb{Z}^5$. After choosing an origin in that coset and dividing coordinates by two, they form another full M -legal carrier tiling. Recognition and coarsening can consequently be repeated indefinitely. At level n , parent anchors lie in one coset of $2^n \mathbb{Z}^5$.

8.5 An actual tiling exists

We construct nested finite patches whose supports exhaust $\mathbb{R}^5$. Put $b_j = 0$ for even j and $b_j = 1$ for odd j , and set

$$a_n = \sum_{j=0}^{n-1} 2^j b_j, \quad t_n = -a_n \mathbf{1}.$$

Subdivide the carrier $2^n P_5 + t_n$ exactly n times using (19). Sibling legality and refinement in Computation 8.3 show that every resulting contact lies in M .

These patches nest exactly. In the next ancestor choose the empty child when $b_n = 0$ and the central child when $b_n = 1$. Both have identity frame, and the chosen child's anchor is

$$t_{n+1} + 2^n b_n \mathbf{1} = t_n.$$

The support contains the cube $[-a_n, 2^n - a_n]^5$. Both endpoints tend to infinity in their respective directions: for example $a_{2m} = 2(4^m - 1)/3$. Thus every bounded region is eventually covered. The union of the nested patches is a full registered M -legal carrier tiling. Complementary profiles across all its shared facets give a physical tiling by CLARK, by Proposition 7.8. Compactness of CLARK follows independently from its definition as a closed bounded polyhedral body.

8.6 No physical tiling has a nonzero period

Suppose a physical tiling $\mathcal{T}$ had a translation period $v \neq 0$. A bounded tile cannot be fixed by any nonzero multiple of v : otherwise it would contain the unbounded orbit of one of its points. We may therefore choose one native frame on each translation orbit of tiles and propagate it by translations through nv . These chosen representatives give the same physical tiling and are periodic as a framed collection.

The registration theorem applies to these choices. After one global Euclidean change of coordinates with linear part L , they become a registered M -legal tiling with period $w = Lv$. Proposition 8.2 makes each parent partition translation-equivariant. Hence w preserves the parent anchors at every level. Since those anchors lie in one coset of $2^n \mathbb{Z}^5$,

$$w \in \bigcap_{n \geq 0} 2^n \mathbb{Z}^5 = \{0\}.$$

This contradicts $v \neq 0$. The conclusion concerns the physical tile collection; preservation of its union alone would be vacuous because that union is all of $\mathbb{R}^5$.

8.7 The full symmetry group of a CLARK tiling

Proof of Theorem 1.4. Let $\mathcal{T}$ be a physical tiling by T_5 , and let

$$\text{Sym}(\mathcal{T}) = \{f \in \text{Isom}(\mathbb{R}^5) : \forall A (A \in \mathcal{T} \iff f(A) \in \mathcal{T})\}.$$

Choose an isometry g_D with $g_D(T_5) = D$ for each tile D , and fix one tile A . The sparse registration theorem applies to arbitrary such choices: for any two tiles D, E , the relative map $g_D^{-1}g_E$ is a signed coordinate permutation followed by an integer translation.

For $f \in \text{Sym}(\mathcal{T})$, put $B = f(A)$. There is a tile $C \neq B$: otherwise the compact tile B would cover all of the noncompact space $\mathbb{R}^5$. Replace the chosen representative of B by fg_A , leaving all other representatives unchanged. Registration for this choice shows that $g_C^{-1}fg_A$ is a signed-integral pose. Registration for the original choices gives the same conclusion for $g_A^{-1}g_C$. Their composition proves

$$g_A^{-1}fg_A(x) = P_f x + t_f, \quad P_f \in B_5, \quad t_f \in \mathbb{Z}^5,$$

where B_5 is the group of signed coordinate permutation matrices. The argument also covers $f(A) = A$, because the auxiliary tile C is distinct from B .

The assignment $f \mapsto P_f$ is injective. Indeed, if $P_f = P_h$, then in the fixed chart g_A the map fh^{-1} is a translation. Conjugating a translation by a Euclidean isometry again gives a translation. Since f and h both preserve $\mathcal{T}$, this translation preserves $\mathcal{T}$. Theorem 1.3 forces its displacement to be zero, so $f = h$. There are $5!$ axis permutations and 2^5 sign choices; hence $|\text{Sym}(\mathcal{T})| \leq 3,840$. This is an upper bound, with no assertion that it is attained. $\square$

8.8 The seven-dimensional conclusion

For HENRY, the sparse registration proof and the exact computation $M_7 = \text{CL}(E_7)$ give a properly framed carrier tiling after one global isometry. Lemma 5.3 makes every such full tiling E_7 -legal, so the registered arithmetic theorem supplies its hierarchy. The exhausting construction in that theorem gives existence. Choosing native representatives coherently along an alleged translation orbit, as above, transfers a physical period to a framed period; the unique hierarchy makes that vector divisible by 2^n for every n , hence zero. Together with closedness and boundedness, this proves the seven-dimensional part of Theorem 1.3. Its exact finite input is separately documented; the complete Lean theorem remains unfinished.

9 Formal verification

The five-dimensional translation-aperiodicity theorem and the full-symmetry theorem use the same physical definition of a tiling: a collection of arbitrary isometric images of the exact closed body, covering $\mathbb{R}^5$ with pairwise disjoint interiors. Translations, rotations and reflections are all allowed. A symmetry must preserve the collection of tiles, not merely its union.

The original CLARK theorem—compactness, tiling existence and absence of every nonzero translational period—has a complete Lean proof and has passed comparison with an independently stated geometric claim, followed by acceptance by Lean’s default kernel, NanoDa and con-ron. The definition and proof links beside Theorem 1.3 identify exactly which body and proposition were checked.

Theorem 1.4, bounding every full symmetry group by 3,840, has a separate complete Lean proof. Both proofs use the standard logical foundations recorded with their formal statements. Their distinct verification histories, exact assumptions, toolchains and records are given in Appendix D.

The seven-dimensional physical result remains computer-assisted, with its complete Lean proof unfinished. The results quantified over prime dimensions remain ordinary mathematical theorems; formalization of individual ingredients does not constitute their complete formal verification.

10 Further questions

Which smaller facet decorations enforce a hierarchy in further dimensions? The five-dimensional right- K quotient gives one way to accommodate a mating relation larger than $\text{CL}(E)$, but each proposed relation requires its own recognition and closure proof.

A separate question is whether finite neighborhoods can characterize the contacts that extend to full tilings. The proof here uses two complete coronas for parent recognition and full coverage for $\text{CL}(E)$ -to- E legality; it does not require a first-corona pruning algorithm to recover E .

The remaining formal targets are the full seven-dimensional physical theorem and a theorem quantified over prime dimensions. The original five-dimensional theorem has completed independent verification.

Author contribution and AI assistance. The research direction and initial conjectures originated with the author. AI assistants across multiple models contributed to the mathematical arguments, computations, Lean proof development and drafting. The author directed the work and takes responsibility for the claims. Formal verification is reported separately for each result.

A Exact data and reproduction

The companion source package contains the two unchanged final JSON specifications in `data/`. Their SHA-256 identities are:

`tile5_final_spec.json`

8c94499b8b9fc89efef794497723f649a188de6a7a5460849dc7989bc61b6094

`tile7_final_spec.json`

2d8a6562eb1ee4f7eb0af571c783d92fb5c3296bc638848aad13ebcd2230a2

In each file the facet records identify the occupied cell, normal axis, normal sign, tangent permutation, tangent signs and signed height of every key. Equation (14) and these records determine all vertices. The five-dimensional file also stores a proposed mate list; the independent verifier reconstructs the geometry first and compares that list only after enumeration.

A.1 The opening section through a CLARK patch

The opening image is a planar section of the level-five carrier patch 2^5P_5 , of side 64. Its plane is

$$x = C_0 + \sigma u + \tau v, \quad u_i = \sqrt{2/5} \cos(2\pi i/5), \quad v_i = \sqrt{2/5} \sin(2\pi i/5), \quad i = 0, \dots, 4,$$

with the supplied offset

$$C_0 = (40.12787310873329, 40.15101405749027, 40.13330076245207, \\ 40.138820527119655, 40.147602655315815).$$

The main panel shows carrier sections; the two circular insets magnify this same affine plane and resolve actual matching boundary keys. Both in-plane directions have the same scale. The magnifications relative to the main panel are $11.6249\dots$ and $688.8852\dots$, displayed as approximately $12\times$ and $690\times$. The tiny target ring in the first inset is enlarged by a factor of 2.2 for visibility. Colour records the number of negative signs in a tile’s signed-coordinate frame; every displayed frame is proper. Heavier lines mark larger carrier supertiles, and the dashed line is the flat carrier-facet reference.

Separate checks reconstruct all 3,630 displayed carrier poses and five matched bump–socket pairs. For each pair all 17 world-coordinate key vertices coincide. The key-vertex equalities use rational arithmetic; plane clipping and rasterization are numerical renderings of the specified geometry. The proof of aperiodicity is the infinite-tiling argument in the main text.

A.2 Other figures and reproducibility

The vector figures use the carrier and reference-key coordinates specified in the paper. Their supplied source is under `figures/scripts/`; the key renderer reads the same T_5 JSON. Independent geometric checks accompany the selected PDFs. In Figure 3(g), source facet 0/key 0 is paired with facet 76/key 0 of $T_5 - \mathbf{1}$ (zero-based indices). The section has

$$x_0 = -z, \quad x_2 = 9/20 - u_0, \quad (x_1, x_3, x_4) = (569/1920, 507/1280, 277/800).$$

No other key enters the displayed local window. For the key panels, at height $z = th$ the tangent box is $to + (1-t)Q_0$. Fixing an omitted $u_j = o_j$ adds only $(1-t)|o_j| \leq (1-t)w_j$, which explains the stated equality of a projection and its specified section. The plots are illustrations of exact data; they add no formal-verification claim.

The package’s `SOURCE_MANIFEST.csv` identifies every retained source by original relative path, packaged path, byte count and SHA-256. `SHA256SUMS` records the artifact bytes. Historical documents are copied unchanged; their earlier status sentences must be interpreted at their own dates, with Section 9 fixing this paper’s formal claims.

The standalone `replay_contacts.py` accepts dimensions five and seven. It loads the independently written verifier with the packaged JSONs and reference catalogs, writes new outputs under `replay_output/`, and does not run Lean. The underlying verifier enumerates a finite complete key-pair candidate set and tests full shared-facet profiles. Its original retained certificates and separate certificate replay are in `evidence/sparse_contacts/`. A certificate replay alone checks its recorded candidates; candidate completeness is supplied by the key-reconstruction argument and the enumerating verifier.

For the ordinary uniform arguments, `evidence/uniform_prime/` includes the complete original proofs, small exact check scripts and outputs. `evidence/dense_registration/` and `evidence/uniform_facet/` contain the separate dense construction and closure proof. The finite $p = 3, 7$ sanity checks for the latter are supplementary evidence; the theorem for every prime in its scope is proved algebraically.

The exact original Lean source checkpoint is supplied as the unchanged archive `evidence/lean/Lean_T5_proved_T7_open_checkpoint_20261006_1345.zip`. Its internal `README.md`, `bootstrap_official.sh`, `run_lean.sh`, version files and trusted-base audit describe the version-pinned environment and a serial rebuild. The wrapper `run_lean.sh` is used as a Lake command wrapper; the source project documents `./run_lean.sh` build for a complete build and `python3 check_module.py SparseMonotiles.T5Monotile` for the final module. Reproducing the import-level audit additionally uses `-t 0`, as recorded in its exact command receipt. Dependencies and official toolchain downloads require network access and substantial memory/disk space. The source archive excludes any assertion that the unfinished seven-dimensional target is proved.

The supported-toolchain source checkpoint is `evidence/lean/Lean_T5_proved_supported_435_checkpoint_20261006_161844.zip`, with SHA-256

9a552795bff26ccdfdf8dee8329acb5f9d2094f887e4e612232b7727c1119165.

It includes the complete Lean 4.35.0-rc2 receipt and the recorded pilot-artifact reuse. Independent checker and acceptance results are outside that receipt’s scope.

The public proof repository is `jconorgrogan/palomar-chair-verification`. The current source is pinned to

9008dec3afc006a62d3d2fd6b49b12f189248f33,

with a downloadable source archive. From the companion package, the one-command reproduction is

`bash reproduce_verification.sh`

It checks out that exact commit and runs the repository’s Linux verification script. It requires Git, Bash, Python, curl, working Bubblewrap namespaces, the pinned prebuilt Lean 4.35.0-rc2 bundle, network access and adequate build resources; `REPRODUCTION.md` states the full prerequisites and scope. The official retry completed successfully; its full comparison and independent-kernel results are recorded in Section D.4. Registry intake and acceptance are separate.

The first external preflight used the immutable source commit

68e28dd1ae9472fc92a8af87b666501c17aa7675.

Its recorded run completed the full solution build and exported the challenge and solution, but the Comparator rejected a mismatch in the helper constant `PalomarMonotiles.tangentIndex._proof_4`. The independent kernel replays were not reached. This result is distinct from the completed local Lean checks above. The source links and hash-bound status record are retained in `evidence/lean/public_preflight_20261006/`; no passing external comparison or registry acceptance is inferred from this run.

To compile this paper, run `bash build.sh` in its root directory with a standard pdfLaTeX installation. No network, Lean build or high-dimensional contact enumeration is performed by the paper build.

Later verification and symmetry extension. The terminal official report for run 37520435323 is retained under `evidence/lean/official_retry_20261006/`. The new full-symmetry proof and its local receipts are under `evidence/lean/finite_symmetry_20261006/`; its separate audit is in the adjacent `finite_symmetry_audit_20261006/` directory. The separate `bash reproduce_strong_verification.sh` command checks out the exact stronger source snapshot and invokes its configured proof/comparison/kernel replay. Its prerequisites and pending official status are given in `REPRODUCTION.md`. The original `reproduce_verification.sh` remains pinned to the independently verified translation-aperiodicity theorem.

The first published version is Zenodo DOI 10.5281/zenodo.23197475. This revision adds the completed independent-verification record and the separate full-symmetry theorem without changing either exact tile specification. The revised front-page artwork states the finite-symmetry theorem and distinguishes its Lean proof from the independently verified translation-aperiodicity theorem.

B Proof and evidence ledger

Claim	Evidence used	Exact scope or remaining limit
Arithmetic frames and dissection	Section 2; original uniform reduction	Every odd prime, square μ , nonsquare g , explicit empty role
Exact E and $\text{CL}(E)$	Sections 3, 5; symbolic and coarse-legality sources	Full poses; all offsets tested; no claim that $E = \text{CL}(E)$
Unique hierarchy and existence	Sections 4, 5; star completion and exhausting ancestors	Full registered framed tilings; ordinary uniform proof
Dense physical family	Section 6; separate registration and full free-orbit closure	Primes $p \equiv 3 \pmod{4}$, $\mu = 1, g = -1$; finite full symmetry; ordinary proof
Sparse registration	Section 7.2; expanded registration proof; exact geometry receipt	Exact final one-height bodies; arbitrary isometries; full profiles; no dense-proof substitution
Sparse M_5, M_7	Computation 8.1; independent exact enumeration and certificate files	284 and 408 accepted poses; all proper; whole-facet matching, including accidental single-facet matches
Five-dimensional hierarchy	Proposition 8.2 and Computation 8.3; parent recognition and exact closure	Right- $\{\text{id}, R\}$ classes; no global 222-pose lift assumed
CLARK tile T_5 physical theorem	Lean 4.19 and 4.35.0-rc2; successful official Comparator, Lean, NanoDa and con-ron checks	Compactness, existence, no nonzero tile-collection translation period; no hypotheses
CLARK full symmetry bound	Separate Lean 4.35.0-rc2 <code>T5_strong_aperiodic_bound</code> ; local compile and semantic audits	At most 3,840 full Euclidean symmetries of any tiling; no additional hypotheses; independent replay pending

Claim	Evidence used	Exact scope or remaining limit
Compact body binding	Lean 4.35.0-rc2 body-equality and claim-equivalence receipts	Both dimensions; equality of definitions alone is not the main monotile theorem
HENRY tile T_7 physical theorem	Ordinary proof plus finite exact M_7 computation	Complete Lean endpoint remains open; no finite-full-symmetry assertion for this exact sparse body

B.1 Principal ordinary sources

The unchanged sources retained in the package are organized as follows.

1. `evidence/uniform_prime/UNIFORM_REDUCTION_PROOF.md`, `HOLE_GRAPH_AND_SYMBOLIC_E.md`, `UNIFORM_STAR_COMPLETION.md`, `UNIFORM_COARSE_LEGALITY.md`, and `NONEMPTINESS_AND_APERIODICITY.md` supply the registered theorem. Later companion proofs supersede open-obligation language in the earliest reduction file.
2. `evidence/dense_registration/REGISTRATION_THEOREM.md` fixes the full-site rational construction; `evidence/uniform_facet/FINAL_THEOREM.md` proves the prime-3-modulo-4 full-profile closure. The registration source's historical list of then-unverified finite applications is not its final application status.
3. `evidence/sparse_original/PROOF.md` and `CHAIN.md` fix the original sparse ordinary proof and finite-computation chain. `evidence/sparse_registration/REGISTRATION_EXPANDED_PROOF.md` supplies the connected-crease and carrier-strip expansions used in this manuscript. No old dense-feature or distinct-height argument is applied to the one-height sparse bodies.
4. `evidence/quotient5/QUOTIENT_HIERARCHY.md` supplies the quotient proof. Its body-independent hierarchy argument applies to the sparse body's identified M_5 and R symmetry through the exact binding recorded in the sparse chain.

C Exact finite computations

The finite computations in the five-dimensional proof are stated here as explicit tests on the rational coordinate data. Their interpretation as statements about arbitrary physical tilings is proved by the registration theorem; their use in infinite hierarchies is proved in the main text.

C.1 The exact finite question left by registration

A registered pose is a map $p(x) = Gx + t$ with $G \in B_d$ and $t \in \mathbb{Z}^d$. We say two carrier facets are *opposed and coincident* if

$$Gb_{f'} + t = b_f, \quad Gn_{f'} = -n_f.$$

For signed coordinate frames these equations identify the entire unit facets, not merely their supporting planes. Define M_d to consist of those poses p for which the two carriers have disjoint interiors, there is at least one opposed coincident facet pair, and (17) holds at every such pair. This is a finite geometric mating relation after the following reduction; it is not defined by membership in a preferred catalog.

Lemma C.1 (Profiles can be compared by finite signatures). *On opposed coincident facets f, f' , with tangent map $m = I_{f'}^\top G^\top I_f$, the following are equivalent:*

1. the complete physical height profiles agree;

2. $v_f(\gamma) = -v_{f'}(m\gamma)$ for every $\gamma \in B_k$;

3. the sets of closed key pyramids coincide after transport, with every bump paired with a dent.

These conditions can be tested by equality of the finite sets of base centres, tangent half-widths, apices and signs.

Proof. Substitute $\gamma' = m\gamma$ in the sum defining $H_{f'}(m\gamma)$. The functions $x \mapsto \kappa(\gamma^{-1}x - \bar{x})$ have pairwise disjoint supports and take the value 1 at $\gamma(\bar{x} + o)$. Evaluating the profile identity at these points gives exactly the coefficient identities in (2); substituting them back proves the converse. This includes every absent site, since its coefficient is zero. The same identities pair the bases and signed apices of the keys and give (3).

Conversely, on a fixed common base plane a pyramid has a unique vertex off that plane, namely its apex; its section in the plane is its base box. Coordinate extrema of the box recover its centre and half-widths. Thus a transported solid-key equality recovers the whole signature. The sign records the side on which the tile has material and must be reversed for the opposed outward normals. No symbolic component name occurs in this comparison. $\square$

Lemma C.2 (An exhaustive candidate set). *Let $\mathcal{K}_d$ be the finite list of native keys, each retaining its facet and tangent frame. For every ordered pair $(f, \gamma), (f', \gamma') \in \mathcal{K}_d$ form*

$$G = [I_f \gamma, -n_f][I_{f'} \gamma', n_{f'}]^T, \quad t = b_f - Gb_{f'}. \quad (21)$$

Keep the distinct poses produced. Every element of M_d occurs in this list. In particular every neighbour in any arbitrary physical tiling occurs in it after registration.

Proof. Both bracketed matrices are orthogonal signed coordinate bases. The formula therefore produces a signed coordinate matrix, takes $n_{f'}$ to $-n_{f'}$, and identifies the facet centres. The integer/half-integer centre argument of Lemma 7.6 proves $t \in \mathbb{Z}^d$. For completeness choose one shared facet of a pose in M_d and one key on that facet. By Lemma C.1 it has a complementary key on the other facet. The actual relative pose satisfies (16), which uniquely solves to (21). Thus it is generated by this ordered pair. Proposition 7.8 supplies the final assertion for physical tilings.

The generator deliberately uses all ordered key pairs, even ones whose two signs do not complement. Such a pair may create an invalid candidate; it cannot remove a valid one. No restriction to proper frames is imposed. $\square$

Here is the full finite test, with its bounds made explicit for CLARK. There are 31 native cells, 160 exposed facets and 256 keys. Run (21) on the $256^2 = 65\,536$ ordered pairs and discard duplicate poses. For each remaining pose $Gx + t$, use the following tests in the indicated order.

1. Write $(Gx)_i = \epsilon_i x_{\pi(i)}$. The lower corner of the image of the unit cell at c is

$$(p_*c)_i = \epsilon_i c_{\pi(i)} + t_i - \frac{1 - \epsilon_i}{2}.$$

Reject if $C_5 \cap p_*C_5 \neq \emptyset$. This is at most 31^2 integer-vector comparisons and is exactly carrier interior overlap, since both carriers are unions of lattice unit cells.

2. Among all 160^2 ordered facet pairs, retain precisely those satisfying $Gb_{f'} + t = b_f$ and $Gn_{f'} = -n_f$. Reject if there are none. This tests full unit-facet contact, not just closeness or intersection of supporting hyperplanes.

3. For every retained pair compare its complete finite signed profiles. A key signature (z, r, a, σ) consists of the base centre, its coordinate half-widths (zero in the normal coordinate), its apex, and its sign. Transport a source signature to

$$(Gz + t, |G|r, Ga + t, -\sigma).$$

Accept only if the resulting set is exactly the root set at every shared facet. Checking both inclusions is essential: an extra source key opposite an empty root site is a rejection.

All quantities are rational; multiplying key coordinates by 19 200 makes every CLARK signature integral. The four nonzero scaled half-widths are 240, 300, 360, 420, the apex's normal height has magnitude 80, and the four tangent offset magnitudes are 80, 75, 72, 70. Facet centres may separately be stored doubled. Thus the test consists of bounded loops and exact integer equalities, with no floating-point tolerances or geometric oracle.

The exhaustive outcome is 36 204 distinct candidate poses: 2020 fail the carrier-overlap test, 33 900 fail the full profile test, and 284 are accepted. These are results of the specified finite comparison, whereas its exhaustiveness and its interpretation as a physical mating law are the lemmas just proved. To identify the accepted set with an arithmetic catalog one must compare the actual poses, not only the cardinality 284. A rejection can be checked from a common occupied cell or from one shared facet and one missing signed key; an acceptance requires all shared-facet equalities. These provide independently checkable witnesses for both outcomes.

The reflected candidates are included in this count. A separate short properness check is available for CLARK: on each of its 256 listed keys,

$$\sigma \det[I_f \gamma, n_f] = 1.$$

This is a finite sign identity, calculated from the row's facet axis, normal side, tangent permutation parity and tangent signs. For a matched pair, the frame identity and the opposite signs give

$$\det G \sigma' \det[I_{f'} \gamma', n_{f'}] = \sigma \det[I_f \gamma, n_f],$$

hence $\det G = 1$. This argument uses the CLARK placement list. It does not apply to the exact T_7 list, whose corresponding signs are not constant; in dimension seven properness must be supplied by the complete mating calculation.

C.2 Finite identities of the five-dimensional contact list

The identities used in the parent-recognition proof can be checked from the 284 accepted poses without any geometry engine. Reconstruct the 32 children from (19); list their occupied unit cells using the signed-coordinate cell formula above. They occupy exactly the 992 cells of $2P_5$, with no repeated owner. For every two adjacent occupied cells with different child owners, compute the owners' relative pose. The 106 unordered contacting sibling pairs give 96 distinct relative sibling poses, all in M .

Conjugating every child by R gives exactly the same 32 poses. Multiplying either endpoint of a contact independently by R leaves M invariant. For each pose, compute its omitted cell and test whether the root owns it; the 62 resulting incoming representatives give exactly the 31 canonical right- K classes. There are also 62 outgoing representatives and 160 wall representatives. Each wall's box lower corner is $\pm 2e_j$, and its omitted cell obeys (18). Every contact translation has constant coordinate parity, and every frame is proper. The short script `evidence/clark_finite_lemmas/check_finite_inputs.py` performs these exact comparisons directly from the pinned list and records their outcomes.

The separate finite equality $R(T_5) = T_5$ is checked on the actual key placement data, including key types and transformed facets; it is not inferred merely from a symmetry of the carrier or of the contact list. The retained body-symmetry result also appears in the formal proof used for the physical theorem.

C.3 Complete parent-contact enumeration

Every child-legal touching pair of canonical parents supplies a witness

$$F = h_r G h_s^{-1}, \quad v = u_r + h_r t - F u_s, \quad r, s \in \{32 \text{ child roles}\}, (G, t) \in M.$$

The 290,816 witnesses give 31,426 distinct parent candidates. Exact occupied-cell or box-minus-corner comparison rejects 27,588 for interior overlap. For each remaining candidate, assign every occupied fine cell to its unique child, and inspect every shared unit face across the two parents. This tests every actual cross-child contact, with no restriction to one facet or one surrounding layer. The root parent has 992 occupied unit cells and 2,511 distinct exterior halo cells available for this search.

Of the 3,838 disjoint candidates, 3,554 fail a child-contact test and 284 survive. Every surviving translation is even, and halving the 284 poses gives exactly M as a set, not just in cardinality. This proves the alignment and both directions of Computation 8.3. The unmodified checker and its original receipt are in `evidence/quotient5/`; the package’s source-layout reconstruction instructions explain how to rerun historical scripts with their retained relative paths.

C.4 The seven-dimensional finite input

The independently implemented seven-dimensional enumeration uses all $1,024^2 = 1,048,576$ ordered key pairs. They yield 692,790 distinct candidate poses, equally split between determinant signs. Of these, 11,198 overlap and 681,184 fail a complete shared-profile comparison; the 408 accepted poses are all proper and equal $CL(E_7)$ exactly.

The requirement to test every shared facet is essential. In dimension seven, 196,608 individual facet triples match accidentally at the single chosen height; they disappear when the complete pose is tested on every shared facet. Thus a one-facet match does not characterize M_7 . The exact full-profile predicate and candidate completeness above cover these cases.

D Formal statements and verification records

D.1 The checked five-dimensional proposition

For the CLARK tile T_5 , the formal model uses $\text{EuclideanSpace}(\mathbb{R}, \text{Fin } d)$ with its Euclidean metric. A tile collection is a set of subsets of this space. A tiling requires that every member be an image of the exact body under an arbitrary isometric equivalence, that the collection cover every point, and that distinct members have disjoint interiors. It does not assume lattice registration, proper frames, hierarchy, or existence.

The period predicate is

$$\text{IsPeriod}(\mathcal{T}, v) \iff \forall A [A \in \mathcal{T} \iff A + v \in \mathcal{T].$$

The target is therefore

$$\text{IsCompact}(T_5) \wedge (\exists \mathcal{T} \text{ IsTiling}(T_5, \mathcal{T})) \wedge (\forall \mathcal{T} \text{ IsTiling}(T_5, \mathcal{T}) \Rightarrow \forall v \text{ IsPeriod}(\mathcal{T}, v) \Rightarrow v = 0). \quad (22)$$

The checked declaration is

`SparseMonotiles.T5Goal_proved : SparseMonotiles.T5Goal.`

It has no hypotheses. The public proof source contains both this declaration and `T5_isAperiodicMonotile`, which combines the checked compactness, existence and physical aperiodicity under arbitrary isometries theorems. Its target definition is available in the same immutable source snapshot. The exact JSON coordinate lists enter through the literal key data and closed-body definition, not through an assumed registration axiom.

D.2 Original toolchain and trusted assumptions

The completed endpoint was checked on Lean 4.19.0 with mathlib v4.19.0. The retained `T5_MONOTILE_RECEIPT.json`, timestamped 6 October 2026 at 13:40:59 UTC, pins 1,320 source/object module pairs. A separate import-level kernel recheck of `T5FinalAudit` completed with exit code zero. Its printed dependencies for the endpoint are exactly

`propext, Classical.choice, Quot.sound.`

These are the reported standard logical dependencies; see the Lean documentation for their meaning [6]. There is no reported `sorryAx`, custom theorem axiom or `native_decide` dependency in this endpoint. Failed exploratory builds are not counted as proofs. A proof-assistant result still relies on the stated logical foundation, implementation and correct interpretation of the formal definitions.

The archived project includes the source, scripts, logs, source-binding records and version pins. The Lean 4.19 result is documented by the original receipt and successful kernel log. The later verification on Lean 4.35.0-rc2 is recorded separately below.

D.3 Verification on Lean 4.35.0-rc2

The completed migration uses Lean 4.35.0-rc2 with mathlib commit

`065356127b1dc0016f66b7283ce0ce2c4055aa55.`

The compact-list/body equalities and corresponding claim equivalences for both dimensions have checked receipts on that toolchain. Those results establish equality between the compact challenge definitions and the literal bodies; they do not alone prove existence or aperiodicity.

The final receipt `T5_MONOTILE_435_RECEIPT.json`, created on 6 October 2026 at 16:18:39 UTC, records the complete five-dimensional theorem and has SHA-256

`bbe75ccd0acb87411c9ab3df43b1cec4a348cd78f229f1439c8a54ce9efe31d9.`

Its import closure contains 1,332 local modules: 1,327 have new checked receipts and five reuse exact previously checked pilot sources and artifacts. No module is recorded as stale or pending. The final `T5FinalAudit` import succeeds; the declarations `T5Goal_proved`, `T5_isAperiodicMonotile`, `T5_hasTiling` and `T5_isAperiodic` have no hypotheses and report only `propext`, `Classical.choice` and `Quot.sound`.

The receipt preserves the pilot provenance: three of the five original pilot receipts did not record per-mathlib dependency hashes, while the final audit checks their current imports. This is a completed Lean kernel result with the stated reuse, not a claim that every module was rebuilt from scratch. The later independent run is recorded below; registry intake and acceptance remain separate. Appendix A preserves both the public source snapshot and the unsuccessful first external comparison.

D.4 Completed independent verification

On 6 October 2026, the full official mechanical preflight at

GitHub Actions run 37520435323

completed successfully for source commit

9008dec3afc006a62d3d2fd6b49b12f189248f33.

The selected declaration was `PalomarMonotiles.T5_isAperiodicMonotile`, the compactness, existence and translation-aperiodicity statement (22). The Comparator accepted its comparison with the independent challenge, with no definition exemptions. Lean’s default kernel, NanoDa and con-rn accepted the proof; con-rn reported 69,348 accepted declarations in verified mode. The permitted axioms were exactly `propext`, `Quot.sound` and `Classical.choice`.

The workflow used the unchanged official pipeline commit

d4e41c1d5b0d114c4859e6e5831dc6d3ad1d0d44

and the `palomar-standard-v1` profile. The verification job completed at 21:13:07 UTC; the terminal report records `status=pass`, `stage=complete`, and empty error and warning lists. Its SHA-256 is

43edc2c785ceaa7ecff4b231a0a9738a5e320d6fd5b0461d6de88491f482f027.

This is a completed mechanical verification of the stated five-dimensional theorem. It does not establish registry intake, editorial acceptance or human peer review. It also does not include the later full-symmetry extension described next.

D.5 The separate Lean proof of finite full symmetry

The additional file `FiniteSymmetry.lean` defines

$$\text{EuclideanSymmetries}(\mathcal{T}) = \{f : \mathbb{R}^5 \simeq_i \mathbb{R}^5 \mid \forall A (A \in \mathcal{T} \iff f(A) \in \mathcal{T})\}.$$

Thus its elements are ambient isometries preserving the physical tile collection, including reflections. The checked declaration `SparseMonotiles.T5_strong_aperiodic_bound` states

$$\text{IsCompact}(T_5) \wedge \text{HasTiling}(T_5) \wedge \forall \mathcal{T}, \text{IsTiling}(T_5, \mathcal{T}) \Rightarrow \begin{cases} \text{Finite}(\text{EuclideanSymmetries}(\mathcal{T})), \\ |\text{EuclideanSymmetries}(\mathcal{T})| \leq 3,840. \end{cases}$$

There are no additional hypotheses. The proof formalizes the fixed-chart registration and injection into signed coordinate permutations in Section 8.7. The source SHA-256 is

a3f3445311f8ffb6b73fdda41065240f652a917138b3e10c3228a36476f4f017.

The new proof terms passed Lean 4.35.0-rc2 locally and in a separate compile audit. All six recorded endpoint and bridge axiom lists contain only `propext`, `Classical.choice` and `Quot.sound`. The audit also checked the intended symmetry predicate and the raw theorem statement. It hash-checked 1,331 imported project sources and 3,993 cached object artifacts against the retained baseline; it did not rebuild that entire import closure. The source, compiler receipts and semantic audit accompany the paper. The stronger statement is now the subject of a separate full official run, using the immutable source snapshot 5f5ceedee770b76c21e8f1105e5e4e3b84b8d676. That run is in progress at this revision’s verification cutoff; no terminal independent result is recorded for it. The completed run in Section D.4 concerns the original translation-aperiodicity statement.

D.6 What remains open in formal verification

The corresponding formal target for HENRY (T_7) is `T7Goal`, with the same physical interpretation. Its complete Lean proof remains unfinished. The seven-dimensional case of Theorem 1.3 uses the written sparse-registration proof and the exact finite 408-pose computation.

The all-prime registered theorem and the dense prime-3-modulo-4 physical family also remain ordinary results. Formal proofs for selected five- and seven-dimensional ingredients do not quantify over every prime. Likewise, externally checked SAT certificates certify the submitted propositional formulas; a mathematical proof is still required that an encoding expresses the intended tiling problem. The geometric and arithmetic proofs above establish the required interpretation.

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